EMERGENCY MEDICINE · CASE WORKBOOK

ER Night Shift

8 shifts · 130 clinical cases

كراسة تدريب مطبوعة — نفس المناوبات الثماني الموجودة في النسخة التفاعلية. حلّ بالقلم، وصحّح من مفتاح الإجابات في آخر الكراسة، وتابع مستواك.



سلّم المستويات — ضع ✓ عند كل مستوى تصل إليه

- ☐ **مستجد** — Starter — البداية
- ☐ **متدرّب** — Junior — بعد المناوبة ١
- ☐ **متمكّن** — Competent — بعد المناوبة ٣
- ☐ **متقدّم** — Advanced — بعد المناوبة ٥
- ☐ **خبير** — Expert — بعد المناوبة ٧
- ☐ **جاهز للتقييم النهائي** — Final-ready — بعد المناوبة ٨
- ☐ **جاهز للشفوي** — Oral-ready — بعد الاختبار الشفوي

1. كل مناوبة = قسم من أقسام طب الطوارئ. ابدأ بأي مناوبة، وأكملها حتى النهاية.
2. الحالات: اختر إجابة واحدة لكل خطوة قبل الانتقال إلى الخطوة التالية.
3. أسئلة الـ Viva: **أجب بصوت مسموع أولًا**، ثم اكتب النقاط.
4. صحّح من المفتاح، وضع ⭕ على كل خطأ وارجع إليه بعد يومين.

زميلكم: أحمد محمود

للإضافات والتعليقات: امسح الباركود أو اضغط عليه

Telegram @A7MDX50 · Based on the Oxford Handbook of Emergency Medicine (Ch. 2–3)

@A7MDX50

SHIFT 1 · BOOK PP. 43–65

Resus & Life Threats

Resus Bay

التركيز على ترتيب الخطوات والجرعات. life threats: anaphylaxis, sepsis, shock, cardiac arrest ;amp& Resus

RESUS 1 · M 28 · BIBA · 14:32 **CASE 1**

Bee sting at a picnic

Stung by a bee 10 min ago. Swollen lips, hoarse voice, wheeze. BP 80/50, HR 128, SpO_2 90%.

STEP 1

Your very first move?

- (A) Request a CXR
- (B) IV chlorphenamine 10 mg
- (C) Take a full allergy history first
- (D) Call for help and assess ABCDE

STEP 2

How do you position him?

- (A) Recovery position
- (B) Head-down tilt 30°
- (C) Lie flat and raise the legs
- (D) Sit him upright at 90°

STEP 3

The drug that saves him — dose and route?

- (A) Adrenaline 500 mcg IM (0.5 mL of 1:1000)
- (B) Adrenaline 300 mcg IV
- (C) Adrenaline 1 mg IV (10 mL of 1:10,000)
- (D) Adrenaline 0.5 mL of 1:10,000 IM

STEP 4

5 minutes later — no improvement. Next?

- (A) Repeat the IM adrenaline
- (B) Switch to oral prednisolone
- (C) Wait 15 more minutes
- (D) Give chlorphenamine instead

STEP 5

He takes bisoprolol and still isn't responding. Add?

- (A) Amiodarone 300 mg
- (B) Calcium chloride 10 mL
- (C) Atropine 500 mcg IV
- (D) Glucagon 1–2 mg IM/IV every 5 min

STEP 6

He settles. Minimum observation?

- (A) At least 4–6 hr after symptoms settle
- (B) 1 hour
- (C) No observation needed
- (D) 30 minutes

💡 ابدأ دايمًا بـ "هنادي على المساعدة وهتعامل بـ ABCDE" — الممتحن بيحب الجملة دي.

Prescribing check **FILL 2**

Anaphylaxis dose grid

Age	Adrenaline IM 1:1000	Chlorphenamine	Hydrocortisone
Adult / child > 12 y			
Child 6–12 y			
Child 6 m – 6 y	150 mcg (0.15 mL)		
Child < 6 m	150 mcg (0.15 mL)		

💡 الهيدروكورتيزون بيتقسم نص كل مرة 200 ← 100 ← 50 ← 25 — والأدرينالين "1.5-5-3-5".

RESUS 2 · F 74 · from care home · 15:05 **CASE 3**

Fever and new confusion

Temp 38.9 °C, RR 24, BP 95/60, newly confused (GCS 13). Dipstick: nitrites +. Lactate 4.3 mmol/L.

STEP 1

Her qSOFA score?

(A) 3
(B) 2
(C) 1
(D) 0

STEP 2

The Sepsis 6 must be done within…

(A) 3 hours
(B) 1 hour of arrival
(C) 24 hours
(D) 6 hours

STEP 3

Initial IV crystalloid volume for her?

(A) 5 mL/kg
(B) 10 mL/kg
(C) 500 mL over 4 hr
(D) 30 mL/kg

STEP 4

When do blood cultures go in?

(A) Not needed with a urinary source
(B) Before antibiotics — only if it causes no delay
(C) Only if the urine culture is negative
(D) After the first antibiotic dose

STEP 5

Still hypotensive after fluids — next?

(A) IV hydrocortisone only
(B) Noradrenaline, target MAP ≥ 65 mmHg
(C) Dobutamine, target SBP 90
(D) Another 3 L of saline

STEP 6

SpO_2 target (no COPD)?

(A) 88–92%
(B) 90–96%
(C) 94–98%
(D) 100%

💡 أهم اتنين بيحسّنوا النتيجة: المضاد الحيوي بدري + المحاليل بدري.

Sepsis protocol **ORDER** 4

The sepsis bundle — in order

اكتب رقم الترتيب في المربع

- [ ] IV crystalloid by response
- [ ] Broad-spectrum IV antibiotics
- [ ] IV access + bloods (FBC, CRP, U&E, LFTs, lactate)
- [ ] Senior / ICU help immediately
- [ ] Blood cultures
- [ ] Vasopressor if still hypotensive
- [ ] Look for the source (consider CXR)
- [ ] ABC + high-flow O_2 as required

💡 رتّبها كأنك بتحكيها للممتحن — الترتيب نفسه هو الإجابة.

MAJORS 6 · F 45 · on chemotherapy · 15:40 **MCQ** 5

Chemo last week, now febrile

T 38.4 °C, feels unwell. Chemotherapy 8 days ago. The FBC isn't back yet. What do you do?

(A) Wait for the neutrophil count before antibiotics
(B) Paracetamol and discharge with safety-netting
(C) Treat as neutropenic sepsis now — e.g. piperacillin–tazobactam 4.5 g IV 8-hourly
(D) Oral co-amoxiclav and review tomorrow

RESUS 3 · M 62 · haematemesis · 16:10 **MCQ** 6

BP 82/50 but HR 64

Haematemesis, BP 82/50, cool and clammy — yet HR is only 64. Most likely reason?

(A) Normal finding in GI bleeding
(B) Septic shock
(C) He is not really shocked
(D) He takes a β-blocker

💡 الفخ الأشهر: مريض على β-blocker مش هيعمل tachycardia.

MAJORS 4 · M 45 · walk-in · 16:25 **MCQ** 7

Clutching his neck while eating

Conscious man, can't speak or cough effectively after a bite of steak. First action?

(A) Start CPR 30:2
(B) 5 back blows, then 5 abdominal thrusts
(C) Blind finger sweep
(D) Encourage him to cough

💡 لو بيكح كويس سيبه يكح — ماتضربهوش على ضهره.

WARD 3 · M 58 · cardiac arrest call · 17:02 **CASE 8**

Collapse on the ward — VF

A 58-year-old collapses. Unresponsive, not breathing normally. Monitor shows VF.

STEP 1

Maximum time for the pulse check?

- (A) 5 seconds per side
- (B) 10 seconds
- (C) 30 seconds
- (D) 1 minute

STEP 2

CPR ratio, depth and rate?

- (A) 30:2 · 5–6 cm · 100–120/min
- (B) 15:2 · 4 cm · 80/min
- (C) 5:1 · 5 cm · 100/min
- (D) 30:2 · 3–4 cm · 140/min

STEP 3

Shock energy (biphasic)?

- (A) 50 J
- (B) 200 J synchronised
- (C) 150 J
- (D) 360 J

STEP 4

Straight after the shock?

- (A) Give adrenaline before anything else
- (B) Check the pulse
- (C) Resume CPR for 2 min immediately
- (D) Look at the monitor for 30 s

STEP 5

Still VF after the 3rd shock. Give?

- (A) Lidocaine 1 mg/kg + amiodarone
- (B) Amiodarone 150 mg only
- (C) Adrenaline 1 mg + amiodarone 300 mg IV
- (D) Atropine 3 mg

STEP 6

Further amiodarone if it keeps recurring?

- (A) 300 mg
- (B) 900 mg
- (C) 150 mg
- (D) None — switch to lidocaine

💡 في الـ shockable التركيز على الصدمة + CPR كويس.

RESUS 1 · M 70 · arrest · 17:40 **CASE** **9**

Pulseless — but the monitor shows a rhythm

No pulse. The monitor shows organised narrow complexes at 70/min.

STEP 1

What is this?

(A) Asystole
(B) Sinus rhythm — patient is fine
(C) VF
(D) PEA

STEP 2

Adrenaline 1 mg — when?

(A) As soon as possible
(B) Only after 10 min of CPR
(C) After the 3rd shock
(D) Never in PEA

STEP 3

Atropine?

(A) No longer routinely recommended
(B) 500 mcg every 3 min
(C) 3 mg IV once
(D) Only if HR < 60

STEP 4

Which is NOT one of the 4 Hs?

(A) Hypertension
(B) Hypovolaemia
(C) Hypothermia
(D) Hypoxia

STEP 5

Which is NOT one of the 4 Ts?

(A) Toxins
(B) Trauma
(C) Tension pneumothorax
(D) Tamponade

💡 الـ 4Hs و 4Ts أهم ٨ كلمات في الوحدة — لازم يطلعوا منك من غير تفكير.

Arrest trolley check **FILL** **10**

Cardiac arrest drugs

Drug	Dose	When
Adrenaline		VF/pVT after 3 shocks · PEA/asystole ASAP
Amiodarone		VF/pVT refractory to 3 shocks
Magnesium sulfate		
Calcium chloride 10%		
Sodium bicarbonate 8.4%		

💡 الكالسيوم والبيكربونات الاتنين ليهم hyperkalaemia — بس الكالسيوم كمان لـ CCB والبيكربونات لـ TCA.

RESUS 1 · post-ROSC · 18:15 **MCQ** **11**

The relatives ask about the pupils

1 hour after ROSC his pupils are fixed. The family asks if he will wake up. You say…

- (A) He will certainly recover
- (B) We will know within 2 hours
- (C) Prognosis is unreliable this early — rarely possible before 24–72 hr
- (D) Fixed pupils mean brain death

Board round question **MCQ** **12**

Central line

Which central vein do you prefer in the ED?

- (A) Right internal jugular, high approach, USS-guided
- (B) External jugular with a long catheter
- (C) Left subclavian, landmark technique
- (D) Femoral for every patient

Key numbers **NUMBERS** **13**

Key numbers — Unit 1

qSOFA respiratory rate	
qSOFA systolic BP	
Lactate → repeat in 2 hr	
Lactate → 30 mL/kg	
Vasopressor MAP target	
Neutropenia threshold	
Oliguria (adult)	
Shock: saline boluses	
Adrenaline IM adult	
EpiPen dose	
Glucagon on β-blocker	
Anaphylaxis observation	
Compression depth / rate	
Biphasic / monophasic shock	
Pads from a pacemaker	
Bicarbonate in arrest — consider if	
Post-ROSC prognosis	
TTM temperature	

Viva — Unit 1

أجب بصوت مسموع أولًا، ثم اكتب النقاط الأساسية

1. Why was NEWS revised to NEWS2?

2. When do you give 30 mL/kg in sepsis?

3. Why might a shocked patient not be tachycardic?

4. Causes of secondary cardiogenic shock?

5. How is hereditary angio-oedema different?

6. What does end-tidal CO_2 tell you during CPR?

7. Complications of central venous access?

8. Describe the Seldinger technique.

✓ أنهيت مناوبة Resus & Life Threats

الصحيح من المحاولة الأولى: _____ / ١٤ حالة · التاريخ: _____

SHIFT 1 DONE

Cardiology

Chest Pain Unit

Chest pain unit: ECG, STEMI, arrhythmias. كثير من الحالات تبدأ من شاشة المونيتور.

ECG review **NUMBERS** 1

ECG basics

Paper speed	
1 small / 1 large square	
Rate from the strip	
Normal axis	
Normal PR	
Normal QRS width	
Normal QTc	
LVH voltage	
Pathological Q	
LBBB pattern	
RBBB pattern	

RESUS 2 · M 55 · BIBA · chest pain · 20:10 **CASE 2**

Central crushing pain

1 hr of central crushing pain, sweaty. SpO_2 97%. ECG: ST elevation in II, III, aVF.

STEP 1

Oxygen?

- (A) 28% Venturi routinely
- (B) Only if SpO_2 < 90% (aim 90–94%)
- (C) Never in ACS
- (D) High-flow 15 L for everyone with chest pain

STEP 2

Which territory / artery?

- (A) Inferior — usually RCA
- (B) Posterior — LAD
- (C) Lateral — circumflex
- (D) Anterior — LAD

STEP 3

Reperfusion of choice?

- (A) Wait for the troponin
- (B) Heparin and admit
- (C) Thrombolysis in the ED for everyone
- (D) Primary PCI — call cardiology now

STEP 4

Antiplatelets?

- (A) Aspirin 300 mg + ticagrelor 180 mg
- (B) Ticagrelor 90 mg alone
- (C) Aspirin 75 mg alone
- (D) Clopidogrel 75 mg + warfarin

STEP 5

He also has ST elevation in V1. Next?

- (A) Ignore — V1 is normal
- (B) Give GTN infusion now
- (C) Record V4R — suspect RV infarct
- (D) Diagnose posterior MI

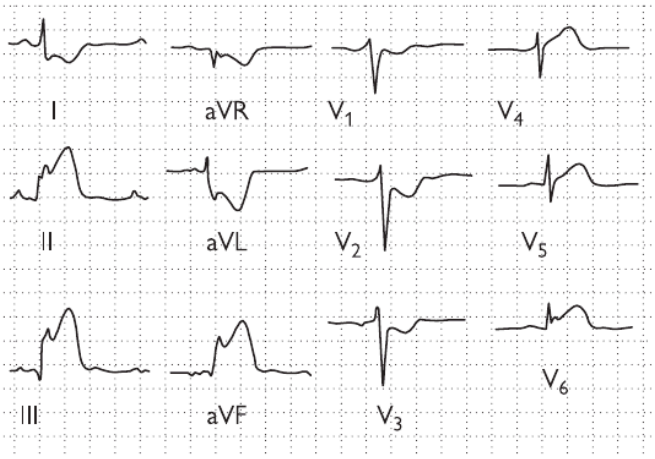
STEP 6

PCI can't happen within 90 min. Then?

- (A) GTN and observe
- (B) Thrombolysis after checking contraindications & consent
- (C) Transfer without treatment
- (D) Wait for PCI however long it takes

💡 لو الـ STEMI سفلي اسأل نفسك على طول: فيه RV؟ — اعمل V4R.

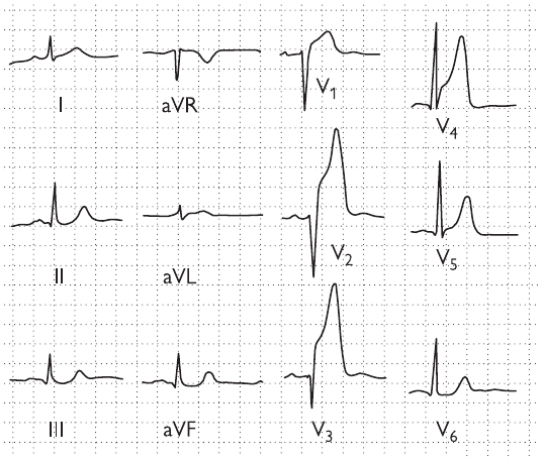
Name this ECG



Which diagnosis fits best?

- (A) Anteroseptal STEMI
- (B) Hyperkalaemia
- (C) Acute inferolateral STEMI
- (D) Acute pericarditis

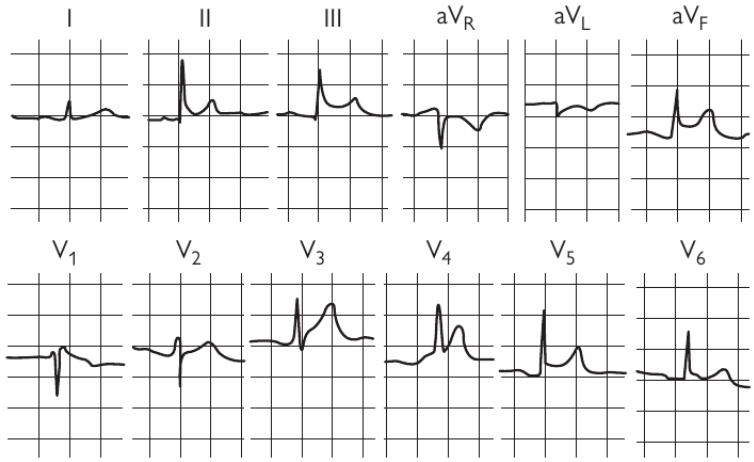
Name this ECG



Which diagnosis fits best?

- (A) Left bundle branch block
- (B) Acute anteroseptal STEMI
- (C) Benign early repolarisation
- (D) Inferior STEMI

Name this ECG



Young man, sharp pain worse lying flat. ECG?

- (A) Inferior STEMI
- (B) Hypokalaemia
- (C) Posterior MI
- (D) Pericarditis — widespread concave ST elevation

Pericarditis or STEMI?

ضع ✓ في العمود الصحيح

	Pericarditis	STEMI
Q waves develop	☐	☐
Concave-up (saddle) ST elevation	☐	☐
Reciprocal ST depression	☐	☐
No Q waves	☐	☐
Electrical alternans with effusion	☐	☐
Hyperacute T waves	☐	☐
PR depression	☐	☐
ST elevation in a coronary territory	☐	☐

💡 ثلاثي البريكاردايتس: ألم + ST + rub ↑ مقعر في كل الليدات.

MAJORS 7 · M 48 · chest pain · 20:55 **MCQ** **7**

Normal ECG, normal troponin

Typical ischaemic pain 2 hr ago. First ECG and baseline troponin are normal. He wants to go home.

- (A) Discharge on aspirin without senior review
- (B) Discharge — ACS excluded
- (C) Do a D-dimer instead
- (D) Don't be reassured — serial troponins (and serial ECGs)

NSTEMI protocol **ORDER** **8**

UA / NSTEMI — in order

اكتب رقم الترتيب في المربع

- ☐ GTN infusion if pain persists (SBP > 90)
- ☐ Fondaparinux 2.5 mg SC
- ☐ O_2 only if SpO_2 < 90% · cardiac monitor
- ☐ Second antiplatelet (clopidogrel 300 / prasugrel 60 / ticagrelor 180 mg)
- ☐ IV opioid ± antiemetic
- ☐ Aspirin 300 mg PO
- ☐ Cardiology if TIMI > 3 or GRACE > 100

💡 التركيبة: مضادين صفائح + مضاد تجلط + GTN لو الألم مستمر.

Pre-thrombolysis checklist **SORT** **9**

Thrombolysis contraindications

ضع ✓ في العمود الصحيح

	Absolute	Relative
Stroke < 3 months	☐	☐
GI bleed < 1 month	☐	☐
Major bleed < 3 months	☐	☐
Traumatic CPR	☐	☐
Platelets < 50	☐	☐
Stroke < 1 year	☐	☐
Subclavian puncture (non-compressible)	☐	☐
Intracranial bleed < 1 year	☐	☐
Any previous intracranial bleed	☐	☐
Pregnancy	☐	☐
BP > 200/120	☐	☐

Bedside calculation **MCQ** **10**

Tenecteplase

Tenecteplase dose for an 85 kg man?

(A) 45 mg single IV bolus over 10 s
(B) 50 mg infusion over 1 hr
(C) 0.9 mg/kg
(D) 30 mg bolus + 30 mg at 30 min

RESUS 3 · M 78 · BIBA · 21:30 **CASE** **11**

HR 34 and dizzy

HR 34, BP 78/40, clammy. ECG: complete heart block with broad QRS.

STEP 1

Adverse features present?

- (A) No
- (B) Can't tell without echo
- (C) Only if he faints
- (D) Yes — shock

STEP 2

First drug?

- (A) Adenosine 6 mg
- (B) Atropine 500 mcg IV
- (C) Adrenaline 1 mg IV
- (D) Amiodarone 300 mg

STEP 3

Minimal response. Interim measure?

- (A) More atropine beyond 3 mg
- (B) Verapamil
- (C) Transcutaneous pacing (or adrenaline 2–10 mcg/min)
- (D) Synchronised DC shock

STEP 4

Definitive treatment?

- (A) Digoxin
- (B) Transvenous pacing
- (C) Oral β-blocker
- (D) Observation

Monitor review **MCQ** **12**

Which bradycardia risks asystole?

Which is NOT a "risk of asystole" feature?

- (A) Mobitz II
- (B) Sinus bradycardia in an athlete
- (C) Ventricular pause > 3 s
- (D) Recent asystole

MAJORS 3 · M 26 · palpitations · 21:50 **CASE** **13**

Narrow regular tachycardia

HR 180, regular narrow QRS. BP 118/76, alert, no chest pain.

STEP 1

Stable or unstable?

- (A) Stable — no adverse features
- (B) Unstable because HR > 150
- (C) Unstable — cardiovert now
- (D) Can't decide without troponin

STEP 2

First manoeuvre?

- (A) Eyeball pressure
- (B) Bilateral carotid massage
- (C) Cold water immersion of the face for 1 min
- (D) Modified Valsalva — blow into 20 mL syringe 15 s, then lie flat legs up

STEP 3

Didn't work. Drug sequence?

- (A) Digoxin 500 mcg
- (B) Adenosine 3 → 6 → 9 mg slow IV
- (C) Adenosine 6 → 12 → 12 mg fast IV bolus + flush
- (D) Amiodarone 300 mg

STEP 4

He is asthmatic. Alternative?

- (A) Flecainide 300 mg PO
- (B) Verapamil 2.5–5 mg IV over 2 min
- (C) Adenosine anyway
- (D) Propranolol

RESUS 2 · monitor review · 22:05 **MCQ** **14**

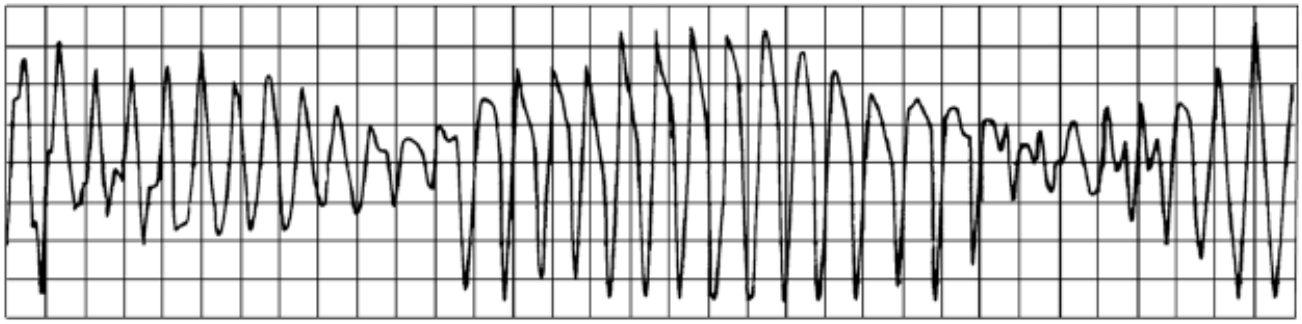
Broad complex, stable

Regular broad complex tachycardia at 170, BP 120/80, alert. Most likely and treatment?

- (A) SVT — adenosine
- (B) Sinus tachycardia — fluids
- (C) VT — amiodarone 300 mg IV over 20–60 min, then 900 mg/24 hr
- (D) AF — rate control with digoxin

RESUS 1 · F 61 · ECG review **ECG** **15**

Name this rhythm



Patient on sotalol, K^+ 3.0. Rhythm and drug?

- (A) Torsades de pointes — $MgSO_4$ 2 g over 10 min
- (B) VT — lidocaine
- (C) VF — amiodarone 300 mg
- (D) AF — digoxin

MAJORS 8 · M 66 · palpitations · 22:40 **CASE** **16**

New AF

Palpitations started 6 hr ago. AF at 140. BP normal, no chest pain, no HF. Known IHD.

STEP 1

Onset < 48 hr, stable. Strategy?

- (A) Immediate DC shock
- (B) Rhythm control
- (C) Nothing — discharge
- (D) Rate control only

STEP 2

He has heart disease. Which drug?

- (A) Flecainide 300 mg PO
- (B) Adenosine 6 mg
- (C) Verapamil
- (D) Amiodarone 300 mg IV

STEP 3

If onset were > 48 hr?

- (A) Cardiovert today
- (B) Aspirin only
- (C) Flecainide IV
- (D) Rate control + anticoagulate

STEP 4

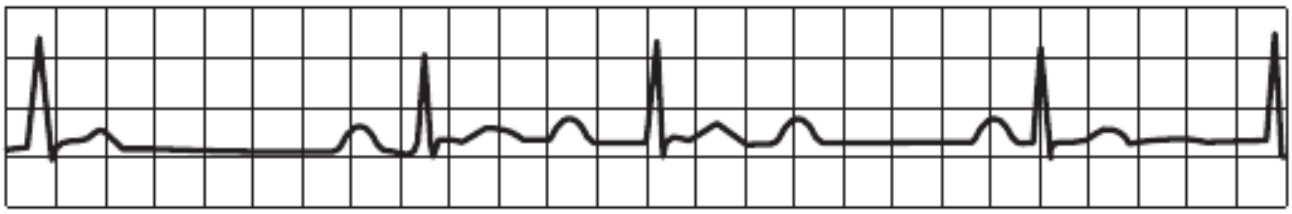
CHA_2DS_2-VASc above which score → consider anticoagulation?

- (A) Any score
- (B) > 1
- (C) > 5
- (D) > 3

💡 أهم رقم: 48 ساعة — بعدها ماتعملش cardioversion من غير anticoagulation.

Rhythm strip **ECG** **17**

Name this rhythm

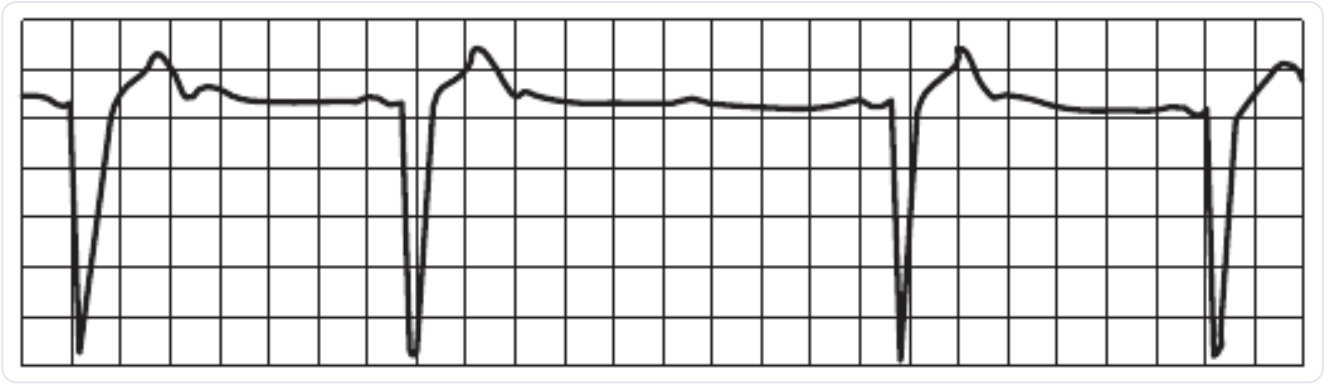


Rhythm?

- (A) Mobitz II
- (B) Complete heart block
- (C) Mobitz I (Wenckebach)
- (D) 1st-degree AV block

Rhythm strip **ECG** **18**

Name this rhythm



Rhythm?

(A) Mobitz II
(B) Sinus arrhythmia
(C) Atrial flutter
(D) Mobitz I (Wenckebach)

Rhythm strip **ECG** **19**

Name this rhythm

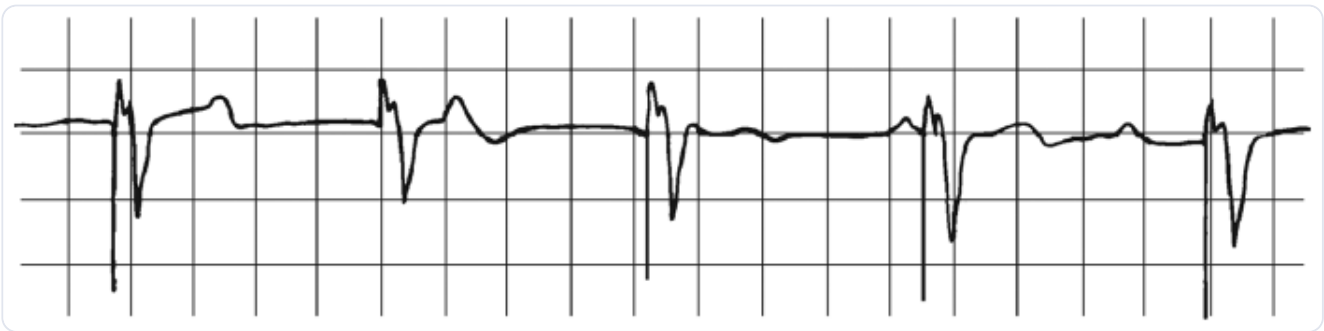


Rhythm?

(A) Complete heart block
(B) 1st-degree AV block
(C) Mobitz I
(D) Sinus bradycardia

Rhythm strip **ECG** **20**

Name this rhythm



Rhythm?

(A) VT
(B) Paced rhythm
(C) Hyperkalaemia
(D) LBBB

RESUS 3 · M 71 · BIBA · 23:15 **MCQ** **21**

Tearing pain to the back

Sudden tearing chest pain radiating to the back, maximal at onset. BP 190/100 right arm, 150/85 left. Next?

(A) CT angiography; labetalol for BP; cardiothoracic early
(B) D-dimer and discharge if normal
(C) Thrombolysis
(D) Aspirin 300 mg + ticagrelor

💡 أي ألم صدر/ظهر مفاجئ وأقصاه من أول لحظة في مريض ضغط = dissection لحد ما يثبت العكس.

MAJORS 11 · M 32 · 23:40 **MCQ** **22**

Cocaine and BP 230/130

Hypertensive emergency after cocaine. Which drug class must you avoid?

(A) Nitrates
(B) Calcium-channel blockers
(C) Benzodiazepines
(D) β-blockers

Key numbers **NUMBERS** **23**

Key numbers — Unit 2

Aspirin in ACS	
Clopidogrel / prasugrel / ticagrelor	
Fondaparinux	
GTN infusion start	
PCI window / lysis deadline	
Alteplase in STEMI	
Reteplase	
Atropine	
Adrenaline infusion (brady)	
Pacing demand rate	
Sync shock: broad / AF	
Sync shock: flutter / SVT	
Amiodarone infusion after load	
Hypertensive emergency BP	
BP reduction first hour	
Keep K^+ post-MI	
Dressler's	

Viva — Unit 2

أجب بصوت مسموع أولًا، ثم اكتب النقاط الأساسية

1. Causes of ST elevation other than MI?

2. Causes of a long QT?

3. Who presents with painless MI?

4. Raised troponin without ACS?

5. How do you diagnose a posterior MI?

6. Sgarbossa criteria?

7. Why avoid amiodarone in torsades?

8. Contraindications to adenosine?

9. CXR signs of aortic dissection?

10. When do you admit pericarditis?

✓ أنهيت مناوبة Cardiology

الصحيح من المحاولة الأولى: _____ / ٢٤ حالة · التاريخ: _____

SHIFT 2 DONE

SHIFT 3 · BOOK PP. 98–125

Respiratory & VTE

Majors — Resp

Respiratory & VTE: لماذا ينقص الأكسجين، وكم نعطي منه؟ — asthma, COPD, pneumothorax, PE.

Oxygen prescribing **SORT** **1**

Who gets which O_2 target?

ضع ✓ في العمود الصحيح

	High-flow (~100%)	90–94%	88–92%
Cluster headache	☐	☐	☐
Sickle-cell crisis	☐	☐	☐
Most acute medical patients	☐	☐	☐
Pneumothorax	☐	☐	☐
CO poisoning	☐	☐	☐
COPD with CO_2 retention	☐	☐	☐
Pneumonia, no lung disease	☐	☐	☐
Type II respiratory failure	☐	☐	☐

💡 الأكسجين دوا — بيتكتب بهدف وجهاز ومعدل. "Can't Cut Supply Please".

Transfer check **MCQ** **2**

Will the cylinder last?

A full E cylinder (680 L) running at 15 L/min lasts about…

(A) 15 minutes
(B) 2 hours
(C) 45 minutes
(D) 10 minutes

RESUS 2 · M 40 · house fire · 00:20 **MCQ** **3**

House fire victim

Smoke inhalation, headache, SpO_2 99% on the probe. Why can't you trust it?

(A) Smoke blocks the light
(B) It is accurate — reassure
(C) Probes are always 2% high
(D) COHb is read as oxyhaemoglobin — check COHb on a VBG

Blood gas review **SORT** 4

Type I or type II failure?

ضع ✓ في العمود الصحيح

	Type I	Type II
Target 88–92%, recheck ABG in 30 min	☐	☐
pO_2 ↓ with normal or low pCO_2	☐	☐
High-concentration O_2 can be life-threatening	☐	☐
pCO_2 ↑ (often ↑ HCO_3^-)	☐	☐
Pneumonia in a previously healthy 30-year-old	☐	☐

MAJORS 4 · F 22 · type 1 diabetes · 00:45 **CASE** 5

Acid–base at the bedside

Na 138, K 5.0, Cl 100, HCO_3^- 10, pH 7.12.

STEP 1

Anion gap = (Na + K) − (Cl + HCO_3)?

(A) 6 — low
(B) 14 — normal
(C) 28 — normal
(D) 33 — high

STEP 2

Commonest cause of a high anion gap acidosis?

(A) Acetazolamide
(B) RTA type 2
(C) Lactic acidosis
(D) Diarrhoea

STEP 3

A raised osmolal gap would point to...

(A) Toxic alcohols — methanol, ethylene glycol
(B) Salicylates
(C) Diarrhoea
(D) Renal failure

MAJORS 6 · F 19 · breathless · 01:05 **MCQ** 6

Panic attack?

Tingling lips, carpopedal spasm, RR 30. "She always does this." Before calling it anxiety you must...

(A) Discharge — known panic attacks
(B) Make her breathe into a paper bag
(C) Exclude PE, DKA, poisoning — SpO_2, ECG, BMG, ABG if not settling
(D) Sedate with diazepam

💡 ماتقولش "ده قلق" إلا بعد ما تستبعد PE و DKA.

Can't finish her sentences

Known asthmatic. PEF 30% of best, SpO_2 90%, RR 30, HR 120. pCO_2 5.2 kPa.

STEP 1

Severity?

- (A) Moderate
- (B) Near-fatal
- (C) Life-threatening
- (D) Acute severe

STEP 2

Why is a normal pCO_2 worrying?

- (A) It isn't — it's reassuring
- (B) It means she has COPD
- (C) It means the ABG was venous
- (D) She should be blowing it off — normal means she's tiring

STEP 3

Bronchodilators?

- (A) Ipratropium alone
- (B) IV salbutamol first-line
- (C) Salbutamol 5 mg + ipratropium 500 mcg, O_2-driven, back-to-back
- (D) Salbutamol 2.5 mg air-driven once

STEP 4

Steroid?

- (A) Inhaled budesonide
- (B) Prednisolone 40–50 mg PO or hydrocortisone 100 mg IV
- (C) Dexamethasone 4 mg
- (D) Not in the acute phase

STEP 5

Poor response — single IV dose of?

- (A) Calcium gluconate
- (B) Aminophylline 5 mg/kg for everyone
- (C) Adrenaline 1 mg IV
- (D) $MgSO_4$ 1.2–2 g over 20 min

STEP 6

Discharge criterion for a less sick patient?

- (A) PEF > 50%
- (B) Stopped wheezing
- (C) PEF > 75% one hour after treatment
- (D) SpO_2 > 90%

💡 الأرقام: 92 / 110 / 25 / 33 / 50 — احفظهم بالترتيب.

Board round question **SORT** **8**

Grade the asthma

ضع ✓ في العمود الصحيح

	Acute severe	Life-threatening
Can't finish sentences	☐	☐
Exhaustion, ↓GCS	☐	☐
RR ≥ 25	☐	☐
Silent chest	☐	☐
PEF 33–50%	☐	☐
HR ≥ 110	☐	☐
SpO_2 < 92%	☐	☐
PEF < 33%	☐	☐
Normal pCO_2	☐	☐

RESUS 3 · M 68 · COPD · 01:50 **CASE 9**

COPD, drowsy

Known COPD, drowsy, flapping tremor. ABG: pH 7.28, pCO_2 8.9, pO_2 7.2 on 28% Venturi.

STEP 1

O_2 target?

- (A) 80–85%
- (B) 100%
- (C) 94–98%
- (D) 88–92%

STEP 2

Nebulisers — driven by?

- (A) Air, with nasal O_2 1–4 L/min
- (B) High-flow O_2 8 L
- (C) Heliox
- (D) Nothing — no nebs in COPD

STEP 3

Steroid?

- (A) Prednisolone 30 mg daily × 7 days
- (B) Prednisolone 40 mg × 5 days
- (C) None
- (D) Hydrocortisone 200 mg × 3 days

STEP 4

Still acidotic after initial treatment. Next?

- (A) CPAP 10
- (B) Aminophylline for all
- (C) Increase O_2 to 60%
- (D) NIV (BiPAP) — CXR first, agree ceiling of care

STEP 5

BiPAP starting settings?

- (A) IPAP 20 / EPAP 10
- (B) IPAP 25 / EPAP 15
- (C) IPAP 10 / EPAP 5 cmH_2O
- (D) IPAP 5 / EPAP 10

💡 قاعدة الضبط: الـ CO_2 عالي ← زوّد IPAP · الأكسجين واطي ← زوّد الاتنين.

fever · 02:15 ;amp& MAJORS 5 · M 72 · cough **CASE 10**

Score the pneumonia

Confused (new), urea 9 mmol/L, RR 32, BP 100/58. Age 72.

STEP 1

CURB-65?

- (A) 4
- (B) 2
- (C) 3
- (D) 5

STEP 2

Meaning?

- (A) Home with amoxicillin
- (B) Severe — admit, IV antibiotics, consider ICU
- (C) Outpatient CXR at 6 weeks
- (D) Admit to a general ward on oral antibiotics

STEP 3

He is septic. Antibiotics?

- (A) Ceftriaxone alone
- (B) Co-amoxiclav 1.2 g IV tds + clarithromycin 500 mg IV bd
- (C) Doxycycline PO
- (D) Amoxicillin 500 mg PO tds

Board round question **MCQ** **11**

After the flu

Which pneumonia can be fulminant after influenza?

- (A) Staphylococcus aureus
- (B) Legionella
- (C) Coxiella
- (D) Mycoplasma

MAJORS 2 · M 24 · chest pain · 02:40 **CASE** **12**

Sudden pleuritic pain

Previously healthy, non-smoker. Breathless. CXR: 3 cm rim at the hilum.

STEP 1

Type?

- (A) Traumatic
- (B) Secondary spontaneous pneumothorax
- (C) Primary spontaneous pneumothorax
- (D) Tension pneumothorax

STEP 2

> 2 cm and breathless. First step?

- (A) Discharge and review in 2 weeks
- (B) Needle aspiration 16–18G, stop at 2.5 L
- (C) Chest drain 28 Fr
- (D) High-flow O_2 alone for 24 hr

STEP 3

Aspiration failed. Next?

- (A) Discharge
- (B) Seldinger chest drain 8–14 Fr and admit
- (C) Surgery today
- (D) Repeat the aspiration

STEP 4

Drain site?

- (A) 2nd ICS mid-clavicular line
- (B) 8th ICS posterior
- (C) 5th ICS, anterior axillary line (safe triangle)
- (D) 3rd ICS parasternal

💡 القرار بيمشي بالأعراض أولًا، وبعدين الحجم 2 سم.

RESUS 2 · ventilated · 03:00 **MCQ** **13**

Gasping, trachea deviated

Ventilated patient, sudden hypotension, trachea deviated to the left, absent breath sounds on the right. Action?

- (A) Urgent CXR first
- (B) CT chest
- (C) Needle decompression 2nd ICS mid-clavicular line, then chest drain
- (D) Increase PEEP

Pleural fluid results **SORT** **14**

Exudate or transudate?

ضع ✓ في العمود الصحيح

	Exudate	Transudate
TB	☐	☐
Pancreatitis	☐	☐
Cardiac failure	☐	☐
Pneumonia	☐	☐
PE with infarction	☐	☐
Hepatic failure	☐	☐
Nephrotic syndrome	☐	☐
Meigs' (ovarian fibroma)	☐	☐
Malignancy	☐	☐

.exudate → upper limit ⅔ ;gt& 0.6 · LDH ;gt& 0.5 · LDH ratio ;gt& Light's: protein ratio 💡

RESUS 3 · F 80 · BIBA · 03:25 **CASE** **15**

Pink frothy sputum

Can't talk, SpO_2 84%, BP 170/100, fine crackles to the apices, cool and sweaty.

STEP 1

First?

- (A) Sit up, high-flow O_2 via tight mask
- (B) Morphine 10 mg IV
- (C) Lie flat
- (D) IV fluids

STEP 2

First-line drug?

- (A) Salbutamol neb
- (B) Diamorphine for all
- (C) GTN for all
- (D) Furosemide 40 mg IV (more if already on it)

STEP 3

Not improving, acidaemic?

- (A) Aminophylline
- (B) Intubate immediately
- (C) CPAP / BiPAP
- (D) More furosemide every 10 min

WARD 11 · F 55 · post-op day 5 · 03:50 **CASE** **16**

Post-op day 5, breathless

Sudden breathlessness. HR 115, SpO_2 91%. You think PE is the most likely diagnosis.

STEP 1

Modified Wells?

- (A) 3 — PE unlikely
- (B) 1.5
- (C) 4 — PE unlikely
- (D) 6 — PE likely

STEP 2

Next test?

- (A) CTPA (no D-dimer)
- (B) Echo
- (C) V/Q only if D-dimer raised
- (D) D-dimer

STEP 3

CTPA delayed 5 hours?

- (A) Thrombolyse
- (B) Aspirin
- (C) Anticoagulate now
- (D) Wait for the scan

STEP 4

She becomes hypotensive. Alteplase dose?

- (A) 50 mg bolus
- (B) 0.9 mg/kg over 1 hr
- (C) 15 mg bolus then 0.75 mg/kg
- (D) 10 mg IV over 1–2 min, then 90 mg over 2 hr

💡 30% من مرضى الـ PE الـ SpO_2 عندهم طبيعي!

Prescribing check **FILL** **17**

Anticoagulants for PE

Drug	Dose
Rivaroxaban	
Apixaban	
Enoxaparin	
Dalteparin	
Unfractionated heparin	

Key numbers — Unit 3

Normal adult RR	
Hypoxia on air	
Hypercarbia	
Anion gap normal	
Osmolal gap normal	
Lactate normal	
Asthma SpO_2 target	
Prednisolone in asthma	
$MgSO_4$ in asthma	
Asthma ventilation	
COPD prednisolone	
CPAP start	
Pneumothorax size cut-off	
Aspiration needle / max volume	
DVT likely (Wells)	
PE likely (Wells)	
Calf difference (Wells)	
Effusion visible / clinical	

Viva — Unit 3

أجب بصوت مسموع أولًا، ثم اكتب النقاط الأساسية

1. When does pulse oximetry lie?

2. Causes of a high anion gap acidosis?

3. ICU criteria in acute asthma?

4. Contraindications to NIV?

5. Mimics of pneumonia?

6. When can you exclude PE with D-dimer?

7. CXR signs of heart failure?

8. Causes of non-cardiogenic pulmonary oedema?

✓ أنهيت مناوبة Respiratory & VTE

الصحيح من المحاولة الأولى: _____ / ١٩ حالة · التاريخ: _____

SHIFT 3 DONE

SHIFT 4 · BOOK PP. 126–131

GI & Liver

Majors — GI

.liver: upper GI bleed, varices, Glasgow–Blatchford, ascites ;amp& GI

RESUS 2 · M 52 · cirrhosis · 04:10 **CASE 1**

Vomiting fresh blood

Known cirrhosis. Large haematemesis, HR 124, BP 88/50, INR 2.1.

STEP 1

Access and bloods?

- (A) One 20G cannula, FBC only
- (B) Central line first
- (C) 2 × 14G, FBC, U&E, clotting, cross-match
- (D) Wait for endoscopy

STEP 2

Vasoactive drug?

- (A) Octreotide SC weekly
- (B) Omeprazole 80 mg bolus
- (C) Adrenaline infusion
- (D) Terlipressin 2 mg IV, repeat every 4–6 hr

STEP 3

Which also reduces mortality?

- (A) IV albumin before endoscopy
- (B) Steroids
- (C) Tranexamic acid
- (D) Prophylactic antibiotics (ciprofloxacin or 2nd/3rd-gen cephalosporin)

STEP 4

Definitive step?

- (A) Emergency endoscopy
- (B) Elective endoscopy tomorrow
- (C) CT angiography first
- (D) Laparotomy

STEP 5

Bleeding won't stop and endoscopy isn't available. Salvage?

- (A) Nothing more
- (B) Repeat terlipressin every 10 min
- (C) Sengstaken / Minnesota tube — gastric balloon first
- (D) Oesophageal balloon to 100 mmHg

.Variceal bleed TAKES a team: Terlipressin · Antibiotics · vitamin K · Endoscopy · Sengstaken 💡

Variceal bleed protocol **ORDER** 2

Variceal bleed — TAKES

اكتب رقم الترتيب في المربع

- ☐ Sengstaken tube if all fails
- ☐ Terlipressin 2 mg IV
- ☐ Endoscopy — emergency
- ☐ vitamin K if INR prolonged
- ☐ Antibiotics (prophylactic)

MAJORS 5 · M 30 · melaena · 04:35 **MCQ** 3

Can he go home?

Glasgow–Blatchford score of 1, no further bleeding. Discharge?

(A) Only if Hb > 100

(B) No — only a score of 0 can go home; > 0 needs urgent endoscopy

(C) Yes, with PPI

(D) Yes — score below 3 is safe

💡 Blatchford = صفر بس هو اللي يروّح — أي رقم أكبر = منظار عاجل.

Bedside calculation **FILL** 4

Glasgow–Blatchford points

Finding	Points
Urea > 25 mmol/L	
Hb < 100 g/L (man or woman)	
SBP < 90	
Syncope	
Pulse ≥ 100	
Melaena	

💡 لاحظ: فيه غلطة مطبعية في جدول الكتاب في درجة الستات (Hb 100–119) — الأصل 1 نقطة.

RESUS 3 · F 79 · PR bleeding · 04:50 **MCQ** 5

Fresh red blood PR, BP 90/60

Commonest source of apparent lower GI bleeding with shock?

(A) An upper GI bleed

(B) Haemorrhoids

(C) Anal fissure

(D) Colorectal cancer

MAJORS 3 · F 58 · PEG tube · 05:10 **MCQ** **6**

The PEG fell out 2 hr ago

Best immediate action?

- (A) Triage ahead, call gastroenterology; keep the track open (e.g. lubricated Foley ≤ original size, don't feed through it)
- (B) Feed through any tube you can pass
- (C) Book an outpatient appointment
- (D) Leave it — reinsert next week

MAJORS 8 · M 47 · alcohol use · 05:25 **MCQ** **7**

Hasn't eaten for a week

Before feeding, which electrolyte is the big danger in refeeding syndrome?

- (A) Chloride
- (B) Calcium
- (C) Sodium
- (D) Phosphate

💡 الخطر: الفوسفات بيقع لما الأكل يرجع.

Board round question **SORT** **8**

Where is the jaundice from?

ضع ✓ في العمود الصحيح

	Pre-hepatic	Intrahepatic	Post-hepatic
Pancreatitis	☐	☐	☐
Paracetamol	☐	☐	☐
Haemolytic anaemia	☐	☐	☐
Gallstones	☐	☐	☐
Pancreatic malignancy	☐	☐	☐
Malaria	☐	☐	☐
NAFLD	☐	☐	☐
Viral hepatitis	☐	☐	☐
Gilbert's	☐	☐	☐

Lab review **MCQ** **9**

Read the LFTs

ALP very high, ALT mildly raised, GGT raised, dark urine, pale stools. Pattern?

- (A) Obstructive (cholestatic)
- (B) Alcoholic hepatitis
- (C) Hepatitic
- (D) Pre-hepatic

MAJORS 6 · M 56 · cirrhosis · 05:50 **CASE** **10**

Cirrhotic with fever and tender ascites

Known cirrhosis, tense ascites, low-grade fever, diffuse tenderness.

STEP 1

Suspect?

- (A) Simple ascites
- (B) Appendicitis
- (C) Spontaneous bacterial peritonitis
- (D) Pancreatitis

STEP 2

What do you send from the tap?

- (A) Nothing — treat empirically
- (B) Glucose only
- (C) MC&S, neutrophil count (EDTA), protein, amylase, cytology
- (D) Lactate

STEP 3

Treatment if severe?

- (A) Amoxicillin PO
- (B) No antibiotics until culture
- (C) Piperacillin–tazobactam 4.5 g IV tds
- (D) Metronidazole alone

Key numbers **NUMBERS** **11**

Key numbers — Unit 4

Blatchford to go home	
Transfusion Hb target in UGIB	
Terlipressin	
Omeprazole (known ulcer)	
Sengstaken oesophageal balloon	
Colorectal share of GI bleeds	
LGIB cross-match	
Refeeding risk	
Visible jaundice	
Refractory ascites 1-yr survival	

Viva — Unit 4

أجب بصوت مسموع أولًا، ثم اكتب النقاط الأساسية

1. Which UGIB patients can be discharged?

2. Drug management of suspected variceal bleed?

3. Bad prognostic factors in UGIB?

4. How do LFTs separate obstructive from hepatic jaundice?

5. Who is at risk of refeeding syndrome?

SHIFT 4 DONE

✓ أنهيت مناوبة GI & Liver

الصحيح من المحاولة الأولى: _____ / ١٢ حالة · التاريخ: _____

Neurology

Stroke & Neuro

.Neurology: dangerous headache, SAH, stroke thrombolysis, status epilepticus

Headache triage **SORT** **1**

Red flag or not?

ضع ✓ في العمود الصحيح

	Red flag	Not a red flag
New headache > 50 y	☐	☐
VP shunt	☐	☐
Known HIV or malignancy	☐	☐
Bilateral band-like, worse with stress	☐	☐
Same pattern as usual migraines for 10 years	☐	☐
Onset during exertion	☐	☐
Sudden onset, worst ever	☐	☐

💡 10–15% من الصداع في الطوارئ وراه سبب خطير.

RESUS 1 · F 34 · sudden headache · 08:10 **CASE** **2**

Thunderclap headache

Sudden severe occipital headache at the gym, peaked within 1 second. Alert, GCS 15.

STEP 1

Ottawa SAH rule — can it rule SAH out here?

- (A) Yes — she is under 40
- (B) The rule only applies over 40
- (C) No — thunderclap and exertion are both positive
- (D) Yes — she is alert

STEP 2

CT timing and sensitivity?

- (A) MRI first
- (B) CT within 6 hr detects > 98%
- (C) CT is useless — go to LP
- (D) CT within 24 hr detects 100%

STEP 3

CT shows SAH. Drug to prevent vasospasm?

- (A) Nimodipine 60 mg PO 4-hourly
- (B) Mannitol 200 mL
- (C) Dexamethasone 4 mg
- (D) Labetalol

STEP 4

Third nerve palsy suggests an aneurysm of?

- (A) Posterior communicating artery
- (B) Basilar tip
- (C) MCA
- (D) Anterior communicating

💡 أي صداع مفاجئ أو "أسوأ صداع في حياتي" = SAH لحد ما يثبت العكس.

Board round question **ORDER** **3**

Hunt & Hess — 1 to 5

اكتب رقم الترتيب في المربع

- ☐ Asymptomatic, mild headache, slight nuchal rigidity
- ☐ Drowsy/confused, mild focal deficit
- ☐ Moderate–severe headache, nuchal rigidity, no deficit except CN palsy
- ☐ Coma, decerebrate posturing
- ☐ Stupor, moderate–severe hemiparesis

MAJORS 4 · F 28 · migraine · 08:35 **MCQ** **4**

Triptan or not?

Who must NOT get sumatriptan?

- (A) Hemiplegic migraine
- (B) Migraine with visual aura in a healthy 25-year-old
- (C) Migraine with vomiting
- (D) Menstrual migraine

MAJORS 6 · F 72 · headache · 08:55 **CASE** **5**

Jaw pain when chewing

New temporal headache, jaw claudication, tender pulseless temporal artery, transient blurred vision.

STEP 1

Diagnosis?

- (A) Migraine
- (B) Trigeminal neuralgia
- (C) Giant cell arteritis
- (D) Tension headache

STEP 2

When do steroids start?

- (A) Immediately — before the biopsy
- (B) Only if vision is lost
- (C) After ESR comes back
- (D) After the biopsy result

Board round question **SORT** **6**

Match the treatment

ضع ✓ في العمود الصحيح

	High-flow O_2 12 L/min	Carbamazepine	Heparin	Blood patch
Cluster headache	☐	☐	☐	☐
Trigeminal neuralgia	☐	☐	☐	☐
Cerebral venous thrombosis	☐	☐	☐	☐
Severe post-dural puncture headache	☐	☐	☐	☐

STROKE CALL · M 68 · BIBA · 09:20 **CASE 7**

Right arm weak, can't speak

Onset 2 hr ago (witnessed). BP 192/108. Weight 80 kg. CT: no bleed.

STEP 1

Within the lysis window?

- (A) No — must be < 3 hr only
- (B) No — must be < 1 hr
- (C) Yes — < 4.5 hr (age 18–80)
- (D) Only if CT perfusion is done

STEP 2

BP before lysis?

- (A) Lysis contraindicated
- (B) Give alteplase now
- (C) Lower to < 180/105 first (e.g. labetalol 10 mg IV)
- (D) Lower to < 140 systolic

STEP 3

Alteplase dose for 80 kg?

- (A) 50 mg over 30 min
- (B) 72 mg — first 7.2 mg as bolus over 1–2 min, rest over 1 hr
- (C) 0.9 mg/kg = 72 mg all as bolus
- (D) 90 mg bolus

STEP 4

Aspirin?

- (A) No antiplatelet for 24 hr until repeat CT is clear
- (B) Clopidogrel 300 mg
- (C) Aspirin 300 mg now
- (D) Aspirin PR

STEP 5

CTA shows a proximal MCA occlusion. Add?

- (A) Mechanical thrombectomy (< 6 hr anterior)
- (B) Heparin infusion
- (C) Nothing more
- (D) Repeat alteplase

💡 الأرقام الذهبية: 4.5 / 3 ساعة للـ 24 / 6 · lysis ساعة للـ thrombectomy.

Pre-thrombolysis checklist **SORT** **8**

Exclusion or not?

ضع ✓ في العمود الصحيح

	Excludes	Doesn't exclude
Woke up with symptoms (unknown onset)	☐	☐
Head injury 6 weeks ago	☐	☐
Type 2 diabetes	☐	☐
Age 70	☐	☐
GI bleed 2 weeks ago	☐	☐
BP 175/100	☐	☐
Any previous ICH	☐	☐
Seizure at onset	☐	☐
Platelets 80	☐	☐

MAJORS 8 · F 81 · stroke · 09:50 **MCQ** **9**

Ischaemic stroke, out of window

CT: no bleed. Not for thrombolysis. Which is correct?

- (A) Swallow screen, then aspirin 300 mg PO/PR ASAP
- (B) Start heparin
- (C) Lower BP to 140 systolic
- (D) Start a statin in the ED

RESUS 3 · M 74 · on apixaban · 10:10 **MCQ** **10**

Intracerebral haemorrhage

ICH on rivaroxaban/apixaban. Reversal agent and BP goal?

- (A) Vitamin K · SBP < 180
- (B) Andexanet · SBP < 140 within 1 hr
- (C) Idarucizumab · SBP < 160
- (D) Protamine · no BP target

MAJORS 10 · M 63 · resolved weakness · 10:30 **MCQ** **11**

TIA — symptoms gone

30-min episode of arm weakness, fully resolved. Correct plan?

- (A) Discharge — it resolved
- (B) CT head now for every TIA
- (C) Aspirin 300 mg now, TIA clinic < 24 hr, statin, no driving until seen
- (D) Calculate ABCD² to decide

Status epilepticus protocol **ORDER** **12**

Status epilepticus — the ladder

اكتب رقم الترتيب في المربع

- [ ] Airway (nasopharyngeal), O_2, monitor
- [ ] Lorazepam 4 mg IV slowly
- [ ] Pabrinex if alcohol/malnutrition
- [ ] IV access + BMG → 50 mL 20% glucose if low
- [ ] Repeat lorazepam after 10 min
- [ ] Phenytoin 20 mg/kg at 50 mg/min with ECG
- [ ] RSI (thiopental) at 30 min

💡 بروتوكول الـ status لازم يتقال بالترتيب والجرعات — ده سؤال شفوي مضمون. "Get Lorazepam, Phenytoin, then RSI."

WAITING ROOM · F 32 · fitting · 10:55 **CASE** **13**

No IV access

Fitting for 7 minutes in the waiting room. Can't get a cannula.

STEP 1

Benzodiazepine without IV?

(A) IM lorazepam 4 mg
(B) Buccal midazolam 10 mg (or rectal diazepam 10–20 mg)
(C) Oral diazepam
(D) Wait for IV

STEP 2

Still fitting, now has IV, 70 kg. Phenytoin?

(A) 2 g push
(B) 100 mg tds
(C) 1400 mg (20 mg/kg) at ≤ 50 mg/min with ECG
(D) 700 mg bolus

STEP 3

She is 32 weeks pregnant. Think?

(A) Eclampsia → $MgSO_4$
(B) Pseudo-seizure
(C) Normal in pregnancy
(D) Hypoglycaemia only

MAJORS 9 · M 29 · first fit · 11:15 **MCQ** **14**

Can he go home?

First fit, fully recovered, normal exam, ECG and electrolytes, with his wife. Plan?

(A) Home: epilepsy clinic within a week, no driving/machinery, supervised bathing — documented
(B) Admit everyone with a first fit
(C) Start valproate in the ED
(D) CT is never needed

RESUS 2 · M 30 · found collapsed · 11:40 **CASE** **15**

Unconscious, pinpoint pupils

GCS 6, RR 6/min, pupils pinpoint. Needle marks on the arms.

STEP 1

Before anything else?

- (A) CT head
- (B) Toxicology screen
- (C) ABC + check BMG
- (D) LP

STEP 2

Diagnostic trial?

- (A) Physostigmine
- (B) Naloxone 0.4–0.8 mg IV
- (C) Atropine
- (D) Flumazenil

STEP 3

Malnourished alcoholic, before glucose give?

- (A) IV thiamine (Pabrinex 2 pairs)
- (B) Nothing
- (C) Oral B12
- (D) Folic acid only

Syncope review **SORT** **16**

Syncope — which need cardiology < 24 hr?

ضع ✓ في العمود الصحيح

	Cardiology < 24 hr	Classic vasovagal — can go
Brother died suddenly at 32	☐	☐
Heart failure	☐	☐
Fainted at the sight of blood, prodrome, quick recovery, normal ECG	☐	☐
New murmur	☐	☐
LOC during exercise	☐	☐
Abnormal ECG	☐	☐

💡 ماتنساش: PE · AAA · ectopic (pregnancy test) · GI bleed (PR).

Neuro exam review **SORT** **17**

UMN or LMN?

ضع ✓ في العمود الصحيح

	UMN	LMN
Increased tone	☐	☐
Upgoing plantars	☐	☐
Reduced reflexes	☐	☐
Wasting	☐	☐
Brisk reflexes	☐	☐
Reduced tone	☐	☐
Fasciculation	☐	☐

MAJORS 12 · M 35 · weakness · 12:05 **MCQ** **18**

Ascending weakness

Weakness climbing up both legs over 3 days, areflexic, after gastroenteritis. What must you monitor?

(A) Hourly glucose
(B) Daily ESR
(C) Serial CK
(D) Serial vital capacity

Key numbers — Unit 5

Ottawa SAH age range	
CT sensitivity for SAH	
LP timing after onset	
Nimodipine	
Fever + headache antibiotic	
Sumatriptan	
Cluster O_2	
Stroke lysis BP limit	
Alteplase stroke	
Platelets exclusion for lysis	
ICH BP target	
Status definition	
Lorazepam in status	
Phenytoin	
Naloxone trial	
Stroke risk after TIA	

Viva — Unit 5

أجب بصوت مسموع أولًا، ثم اكتب النقاط الأساسية

1. When does a normal CT not exclude SAH, and what next?

2. Why give nimodipine?

3. Name five exclusions to stroke thrombolysis.

4. Stroke mimics?

5. First-line drug in status and dose?

6. Causes of delirium?

7. A bedside test for myasthenic ptosis?

8. What is transient global amnesia?

✓ أنهيت مناوبة Neurology

الصحيح من المحاولة الأولى: _____ / ٢٠ حالة · التاريخ: _____

SHIFT 5 DONE

SHIFT 6 · BOOK PP. 158–173

Endocrine & Electrolytes

Majors — Metabolic

جرعات وأرقام. — electrolytes: DKA, hyperkalaemia, sodium, endocrine crises ;amp& Endocrine

RESUS 3 · M 65 · diabetic · 13:10 **CASE 1**

Found confused and sweaty

On gliclazide. Confused, sweaty, GCS 12. BMG 1.9 mmol/L. Not safe to swallow.

STEP 1

Preferred treatment?

- (A) Glucose 10% 50 mL IV, repeat every 1–2 min to max 250 mL
- (B) Glucagon 1 mg IM
- (C) Oral glucose gel
- (D) Glucose 50% 100 mL

STEP 2

Why is glucagon a poor choice for him?

- (A) It causes hypokalaemia
- (B) It is fine — first-line
- (C) It only works IV
- (D) Sulfonylurea hypoglycaemia (also liver failure, chronic alcohol)

STEP 3

He wakes, but it's a sulfonylurea. Watch for?

- (A) Recurrence — glucose infusion may be needed ≥ 24 hr; octreotide
- (B) Hyperglycaemia
- (C) Nothing — discharge now
- (D) Hypernatraemia

STEP 4

If he had NOT woken after correction?

- (A) Wait 6 hr
- (B) Another cause (stroke) or hypoglycaemic cerebral oedema — keep glucose 7–11, ICU, CT
- (C) Discharge
- (D) Give more 50% glucose

💡 اعمل BMG لأي حد في غيبوبة أو تصرفات غريبة أو أعراض عصبية — حتى لو شكله stroke.

Vomiting, deep sighing breaths

Abdominal pain, Kussmaul breathing, BMG 28, ketones 5.1, pH 7.1, K^+ 4.8. Weight 70 kg.

STEP 1

First?

(A) Sodium bicarbonate
(B) KCl 40 mmol stat
(C) Insulin bolus 10 U
(D) 0.9% saline 1 L over 30–60 min

STEP 2

Then?

(A) Sliding scale only
(B) No insulin until K^+ known
(C) Fixed-rate insulin 0.1 U/kg/hr (~7 U/hr) once fluids are running
(D) Insulin 1 U/kg/hr

STEP 3

K^+ 4.8 — potassium?

(A) None — K^+ is normal
(B) Only after 6 hr
(C) KCl 20 mmol/hr (unless $K^+ > 5.5$), hourly K^+
(D) 40 mmol/hr

STEP 4

Glucose falls to 12. Next?

(A) Give 50% glucose
(B) Stop the insulin
(C) Halve the saline
(D) Add 10% glucose 125 mL/hr — keep the insulin

STEP 5

Stop the insulin infusion when?

(A) Bicarbonate > 15
(B) After 12 hr
(C) Glucose < 10
(D) Ketones < 0.3 and pH > 7.3

STEP 6

Precipitants — the 4 I's?

(A) Infection, Injury, Inflammation, Ileus
(B) Infection, Infarction, Insufficient insulin, Intercurrent illness
(C) Infarction, Intoxication, Immobility, Iatrogenic
(D) Insulin, Ischaemia, Iron, Iodine

💡 ماتديش إنسولين قبل المحاليل — وراقب البوتاسيوم كل ساعة.

Board round question **SORT** 3

DKA or HHS?

ضع ✓ في العمود الصحيح

	DKA	HHS
Insulin only if ketonaemia or glucose won't fall	☐	☐
Younger, 1–3 days	☐	☐
Elderly, days–weeks	☐	☐
Altered GCS, reversible focal deficits	☐	☐
Ketonuria	☐	☐
Glucose > 30, ↑ osmolality	☐	☐
Abdominal pain mimicking surgical abdomen	☐	☐
Kussmaul + acetone breath	☐	☐

RESUS 2 · F 30 · marathon runner · 13:55 **MCQ** 4

Na 116, seizing

Acute hyponatraemia after a marathon, now seizing. Treatment?

(A) Tolvaptan
(B) 2.7% saline up to 200 mL over 30 min, recheck
(C) 0.9% saline 2 L fast
(D) Fluid restriction only

Bedside calculation **MCQ** 5

Speed limits

Maximum correction rates?

(A) Hyponatraemia ≤ 1 per hr · hypernatraemia ≤ 10 per 24 hr
(B) No limit if symptomatic
(C) Both ≤ 20 mmol/L per 24 hr
(D) Chronic hyponatraemia ≤ 10 mmol/L per 24 hr · hypernatraemia ≤ 1 mmol/L per hr

💡 الصوديوم المزمن يتصلّح ببطء — وإلا myelinolysis أو cerebral oedema.

Bedside calculation **MCQ** 6

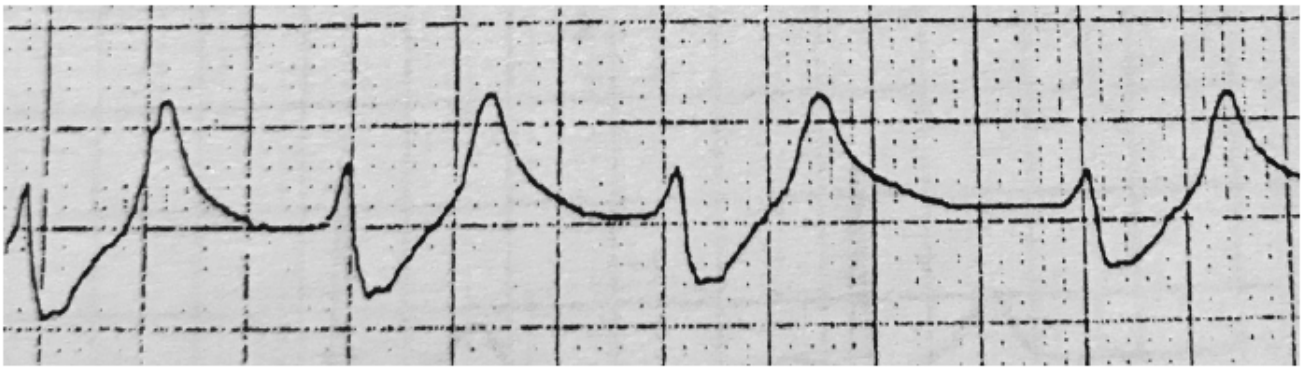
Free water deficit

70 kg, Na 154. Free water deficit = 0.6 × wt × [(Na/140) − 1] ≈ ?

(A) 0.6 L
(B) 9.8 L
(C) 4.2 L
(D) 1.4 L

RESUS 1 · M 58 · dialysis · 14:20 **ECG** 7

K^+ is 7.4



Dialysis patient missed a session. ECG?

- (A) Severe hyperkalaemia — peaked T, absent P, wide QRS
- (B) Torsades
- (C) Hypokalaemia
- (D) Pericarditis

Hyperkalaemia protocol **ORDER** 8

Hyperkalaemia — treatment order

اكتب رقم الترتيب في المربع

- ☐ Bicarbonate if acidotic — shift
- ☐ Salbutamol 5 mg neb — shift
- ☐ Dialysis / treat the cause — remove
- ☐ Insulin 10 U + 50 mL 50% glucose — shift
- ☐ Calcium chloride 10% 10 mL IV — protect the heart

💡 "Cardiac Injury Sometimes Brings Death" — والكالسيوم ما بينزّلش البوتاسيوم، بيحمي القلب بس.

Prescribing check **FILL** 9

Hyperkalaemia doses

Drug	Dose	Effect
Calcium chloride 10%		
Actrapid + glucose		
Salbutamol		Acts in ~30 min
Furosemide (moderate)		Removes K^+

MAJORS 8 · M 77 · on digoxin · 14:45 **MCQ** **10**

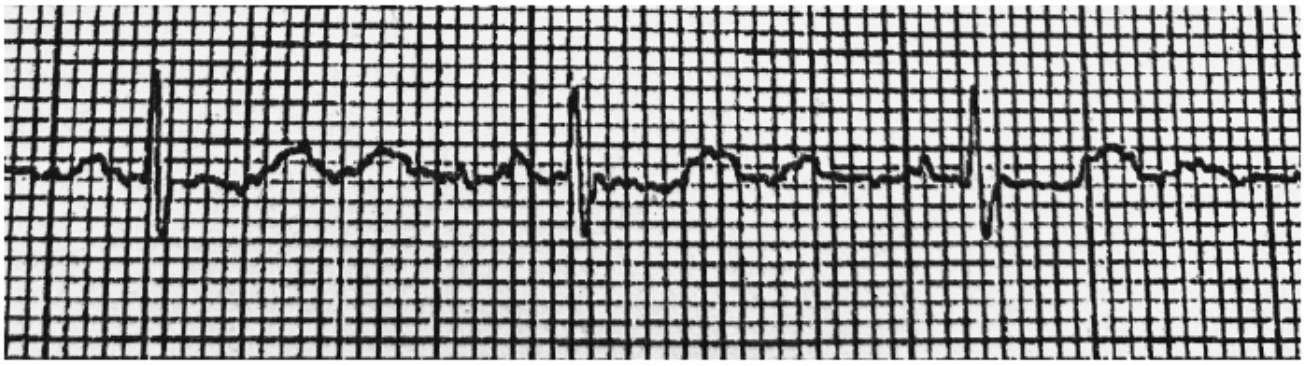
Calcium and digoxin

Hyperkalaemia with ECG changes in a patient on digoxin. How do you give the calcium?

- (A) Don't give calcium at all
- (B) Orally
- (C) Rapid bolus
- (D) Over 30 min — hypercalcaemia potentiates digoxin toxicity

MAJORS 5 · F 66 · weakness · 15:00 **ECG** **11**

K^+ is 2.4



What do you see and how fast do you replace?

- (A) Prominent U waves — max 20 mmol/hr IV with cardiac monitoring
- (B) Delta waves — adenosine
- (C) Peaked T — calcium
- (D) Long QT — magnesium only

RESUS 3 · F 40 · on steroids · 15:20 **CASE** **12**

Shocked after stopping steroids

Stopped prednisolone suddenly last week. BP 76/40, confused. Na 124, K 6.1, glucose 2.8.

STEP 1

Diagnosis?

- (A) DKA
- (B) Septic shock only
- (C) Addisonian crisis
- (D) SIADH

STEP 2

Give now — before results?

- (A) Wait for cortisol
- (B) Dexamethasone 4 mg PO
- (C) Fludrocortisone PO
- (D) Hydrocortisone 100 mg IV stat + 0.9% saline

STEP 3

Glucose 2.8?

- (A) Nothing
- (B) Glucagon
- (C) Insulin
- (D) 50 mL 10% glucose

15:45 · ’RESUS 2 · F 36 · Graves **MCQ** **13**

Thyroid storm

Fever 40 °C, AF 160, agitated, known Graves'. Which drug is contraindicated?

(A) Diazepam
(B) Hydrocortisone
(C) Propranolol
(D) Aspirin — displaces thyroxine from TBG

MAJORS 11 · F 24 · abdominal pain · 16:05 **MCQ** **14**

Abdominal pain, dark urine on standing

Young woman on new OCP: severe abdominal pain, soft abdomen, tachycardia, Na 126, urine darkens on standing. Fits — which drug first?

(A) Phenobarbital
(B) Phenytoin
(C) IV diazepam
(D) Carbamazepine

Key numbers **NUMBERS** **15**

Key numbers — Unit 6

Normal glucose	
Treat hypoglycaemia below	
Glucagon	
DKA first saline	
DKA insulin	
DKA: add 10% glucose when	
DKA KCl	
HHS glucose	
Hypertonic saline	
Hyperkalaemia urgent	
Hypokalaemia severe	
Max K^+ infusion	
Addison / storm hydrocortisone	
Storm propranolol	

Viva — Unit 6

أجب بصوت مسموع أولًا، ثم اكتب النقاط الأساسية

1. Why give potassium when serum K^+ is normal in DKA?

2. When do you stop the insulin infusion in DKA?

3. Does calcium chloride lower potassium?

4. Causes of hyperkalaemia?

5. Commonest cause of an Addisonian crisis?

6. Bloods in Addisonian crisis?

✓ أنهيت مناوبة Endocrine & Electrolytes

الصحيح من المحاولة الأولى: _____ / ١٦ حالة · التاريخ: _____

SHIFT 6 DONE

SHIFT 7 · BOOK PP. 165–169

Renal

Renal Bay

Renal: AKI criteria, dialysis patients, UTI management حسب المريض.

Lab review **SORT 1**

Does it meet AKI criteria?

ضع ✓ في العمود الصحيح

	AKI	Not (yet) AKI
Urine 0.4 mL/kg/hr for 3 hr	☐	☐
Creatinine ↑ 30 µmol/L in 36 hr	☐	☐
Creatinine ↑ 15 µmol/L in 48 hr	☐	☐
Urine 0.3 mL/kg/hr for 8 hr	☐	☐
Creatinine ↑ 60% in 5 days	☐	☐

💡 لو مش عارف هو AKI ولا CKD بيسوء — عامله كـ AKI.

Board round question **SORT 2**

Pre-renal, renal or post-renal?

ضع ✓ في العمود الصحيح

	Pre-renal	Renal	Post-renal
Cardiogenic shock	☐	☐	☐
Sepsis	☐	☐	☐
Bleeding / burns	☐	☐	☐
Glomerulonephritis	☐	☐	☐
Stones	☐	☐	☐
Retroperitoneal fibrosis	☐	☐	☐
Prostate	☐	☐	☐
Acute tubular necrosis	☐	☐	☐
Interstitial nephritis	☐	☐	☐

MAJORS 2 · M 78 · low urine output · 18:10 **CASE** **3**

Not passing urine

On ibuprofen and ramipril, 3 days of D&V. Creatinine doubled. Little urine.

STEP 1

First bedside test for low urine output?

- (A) Kidney USS tomorrow
- (B) CT KUB
- (C) Bladder scan / catheter (flush if already catheterised)
- (D) Renal biopsy

STEP 2

Which test for hyperkalaemia risk right now?

- (A) ECG
- (B) Echo
- (C) CXR
- (D) Urine culture

STEP 3

His drugs?

- (A) Stop NSAIDs and ACE inhibitors
- (B) Double the ACE inhibitor
- (C) Stop only ibuprofen
- (D) Continue both

Board round question **ORDER** **4**

AEIOU — indications for dialysis

اكتب رقم الترتيب في المربع

- ☐ Uraemia (severe symptoms)
- ☐ Electrolytes — intractable hyperkalaemia
- ☐ Acidosis (severe)
- ☐ Intoxication (toxin removal)
- ☐ Overload (fluid)

RESUS 2 · M 60 · haemodialysis · 18:40 **MCQ** **5**

Anuric, in pulmonary oedema

Haemodialysis patient, missed his session, SpO_2 86%, crackles to the apices. What helps?

- (A) Urgent dialysis; meanwhile O_2 + nitrates
- (B) IV fluids
- (C) Furosemide 120 mg IV
- (D) Morphine and wait

MAJORS 4 · M 60 · haemodialysis · 19:00 **MCQ** **6**

The fistula thrill is gone

The AV fistula has no thrill this morning. Meaning?

- (A) Shunt thrombosis — vascular emergency
- (B) Infection
- (C) Normal after dialysis
- (D) Dehydration only — oral fluids

MAJORS 6 · F 55 · peritoneal dialysis · 19:20 **MCQ** **7**

Cloudy PD bag

CAPD patient: cloudy bag, abdominal pain. Gram stain shows several organisms. Suggests?

- (A) Contamination — ignore
- (B) Typical staph peritonitis
- (C) A surgical cause such as a diverticular abscess
- (D) Viral peritonitis

MAJORS 7 · F 29 · 8 weeks pregnant · 19:45 **CASE** **8**

Dysuria in pregnancy

8 weeks pregnant. Dysuria and frequency. Dipstick: leucocytes + nitrites.

STEP 1

Antibiotic?

- (A) Amoxicillin 250 mg tds or cefalexin 500 mg tds × 7 days
- (B) Ciprofloxacin
- (C) Trimethoprim 200 mg bd × 3 days
- (D) Nitrofurantoin × 3 days

STEP 2

Asymptomatic bacteriuria in pregnancy?

- (A) Ignore it
- (B) Treat it
- (C) Treat only if fever
- (D) Recheck in 6 months

💡 الحامل: trimethoprim ممنوع في الأول، nitrofurantoin ممنوع في الآخر.

Prescribing check **SORT** **9**

How long?

ضع ✓ في العمود الصحيح

	3 days	5–10 days	7 days
Pregnant woman	☐	☐	☐
Uncomplicated cystitis in a woman	☐	☐	☐
Man with UTI	☐	☐	☐
Woman with renal impairment	☐	☐	☐
Immunosuppressed woman	☐	☐	☐

MAJORS 9 · F 84 · from care home · 20:05 **MCQ** **10**

Positive dipstick, no symptoms

Catheterised, no symptoms, dipstick positive. Antibiotics?

- (A) Yes — IV gentamicin
- (B) No — asymptomatic bacteriuria in the elderly/catheterised isn't treated
- (C) Yes — change catheter and give 7 days
- (D) Yes — 3 days trimethoprim

Key numbers — Unit 7

AKI creatinine (48 hr)	
AKI creatinine (7 days)	
AKI urine output	
CKD problems start at GFR	
UTI threshold	
E. coli share of UTI	
Pyelonephritis temperature	
Trimethoprim	
Nitrofurantoin	

Viva — Unit 7

أجب بصوت مسموع أولًا، ثم اكتب النقاط الأساسية

1. AKI diagnostic criteria?

2. Indications for dialysis?

3. Signs of transplant rejection?

4. Antibiotics to avoid in pregnancy?

✓ أنهيت مناوبة Renal

الصحيح من المحاولة الأولى: _____ / ١٢ حالة · التاريخ: _____

SHIFT 7 DONE

SHIFT 8 · BOOK PP. 174–185

Haematology

Resus — Bleeding

Haematology: anticoagulant reversal, massive transfusion, transfusion reactions, sickle crisis. بعدها
يُفتح الاختبار الشفوي.

Board round question **SORT** 1

Platelets or clotting factors?

ضع ✓ في العمود الصحيح

	Platelet problem	Coagulation factor problem
Epistaxis, menorrhagia	☐	☐
Bleeding into joints and muscles	☐	☐
Bleeds 2–3 days after surgery	☐	☐
Bleeds immediately after dental extraction	☐	☐
Petechiae and purpura	☐	☐
Retroperitoneal bleed	☐	☐

💡 كلّم الهيماتولوجي في أي مريض عنده مشكلة نزف معروفة أو مشكوك فيها.

Prescribing check **FILL** 2

Reverse the anticoagulant

Drug	Reverse with
Warfarin	
Dabigatran	
Rivaroxaban / apixaban	
LMWH (within 24 hr)	

💡 المضاد فيه حرف الدوا: Xa–andeXanet · Dabigatran–iDarucizumab.

MAJORS 3 · M 70 · on warfarin · 21:10 **MCQ** **3**

INR 8.2, no bleeding

Well, no bleeding, INR 8.2. Plan?

- (A) Vitamin K 10 mg IV and discharge
- (B) Continue warfarin
- (C) Withhold warfarin; specialist advice before vitamin K_1 5 mg slow IV or PO
- (D) PCC immediately

Board round question **MCQ** **4**

Starting warfarin

Why not start warfarin alone for a DVT?

- (A) It needs a loading dose of vitamin K
- (B) It is initially prothrombotic — cover with heparin/LMWH
- (C) It is contraindicated in DVT
- (D) It takes 2 weeks to be absorbed

RESUS 1 · M 30 · major trauma · 21:35 **CASE** **5**

Bleeding out

Pelvic fracture, BP 70/40, HR 140. Injured 1 hr ago.

STEP 1

Definition of massive blood loss in the ED?

- (A) 1 L in 24 hr
- (B) 10 units in 1 hr
- (C) 50% of circulating volume within 3 hr
- (D) Hb < 70

STEP 2

Drug within 3 hr?

- (A) Factor VIIa
- (B) Protamine
- (C) Vitamin K
- (D) Tranexamic acid 1 g IV

STEP 3

Ratio?

- (A) 1 FFP : 4 red cells
- (B) 2 FFP : 1 red cell
- (C) Red cells only
- (D) 1 U FFP : 1 U red cells

STEP 4

Give platelets if below?

- (A) 150 × 10^9/L
- (B) 75 × 10^9/L
- (C) 10 × 10^9/L
- (D) 20 × 10^9/L

STEP 5

Repeat bloods (incl. Ca, Mg, fibrinogen) every?

- (A) 4 hr
- (B) 10 min
- (C) 30 min
- (D) 2 hr

STEP 6

Use a blood warmer when?

- (A) Never
- (B) Always
- (C) Only in children
- (D) > 50 mL/kg/hr (children 15 mL/kg/hr)

Blood bank call **FILL** 6

What to request — and how long

Request	Time
Group & screen (electronic issue)	
ABO/Rh-compatible	
Full cross-match	
O Rh-negative	

WARD 5 · F 52 · during transfusion · 22:00 **ORDER** 7

Suspected transfusion reaction

اكتب رقم الترتيب في المربع

- [ ] STOP the transfusion, keep the line open with 0.9% saline
- [ ] Bloods: blood bank sample, U&E, LFT, IgA, clotting (tryptase if anaphylaxis)
- [ ] Phone blood bank + haematologist
- [ ] Send bag + giving set to blood bank
- [ ] Fluid balance, urine output, urine Hb
- [ ] Obs, O_2, re-check unit label vs wristband

💡 أغلب كوارث النقل غلطة كتابة.

WARD 6 · F 88 · post-transfusion · 22:20 **MCQ** 8

Breathless after 2 units

New pulmonary oedema 3 hr after a transfusion. The biggest killer in transfusion is…

(A) Hepatitis B
(B) Bacterial contamination
(C) TACO — transfusion-associated circulatory overload
(D) ABO incompatibility

Lab review **MCQ** 9

Read the clotting

Septic patient oozing from every line: platelets ↓, fibrinogen ↓, INR ↑, APTT ↑, FDPs ↑.

(A) Haemophilia A
(B) ITP
(C) Warfarin overdose
(D) DIC — treat the cause, expert-guided products

Board round question **SORT** **10**

Which product?

ضع ✓ في العمود الصحيح

	Cryoprecipitate	PCC	FFP
VIII, fibrinogen, vWF	☐	☐	☐
II, VII, IX, X — reverses warfarin	☐	☐	☐
All clotting factors + fibrinogen	☐	☐	☐

MAJORS 8 · M 19 · sickle cell · 22:50 **CASE** **11**

Sickle crisis

Severe bone pain in a cold week. SpO_2 95%. Now develops chest pain, SpO_2 89%, new infiltrate on CXR.

STEP 1

Initial crisis care?

- (A) Fluid restriction
- (B) Ice packs
- (C) Paracetamol and discharge
- (D) Warmth, O_2 if < 94%, IV opioids, hydration, expert help

STEP 2

Chest pain + hypoxia + new infiltrate =

- (A) Anxiety
- (B) PE only
- (C) Simple pneumonia
- (D) Acute chest syndrome — leading cause of death

STEP 3

Classic organism in sickle osteomyelitis?

- (A) E. coli
- (B) Salmonella
- (C) Pseudomonas
- (D) Strep. pyogenes

💡 الأزمة: دفّي + أكسجين + مسكّن قوي + سوائل — والـ acute chest syndrome هو القاتل الأول.

Key numbers — Unit 8

Thrombocytopenia / spontaneous bleeding	
Discuss with haem on anticoagulant if platelets	
Tranexamic acid	
FFP : red cells	
Fibrinogen → cryo	
INR/APTT target	
Red cells rise	
Red cells / TACO-risk rate	
Platelets, FFP, cryo	
Blood storage	
Sickle O_2 threshold	
HbS: trait / SS	

Viva — Unit 8

أجب بصوت مسموع أولًا، ثم اكتب النقاط الأساسية

1. How does bleeding differ in platelet vs factor disorders?

2. Lab picture of DIC?

3. How do you reverse warfarin in a major bleed?

4. Ratio and targets in massive transfusion?

5. Complications of massive transfusion?

6. Commonest cause of death in sickle-cell anaemia?

SHIFT 8 DONE

✓ أنهيت مناوبة Haematology

الصحيح من المحاولة الأولى: _____ / ١٣ حالة · التاريخ: _____

🔑 مفتاح الإجابات

Resus & Life Threats · المناوبة ١

1. Bee sting at a picnic	1-D · 2-C · 3-A · 4-A · 5-D · 6-A
2. Anaphylaxis dose grid	Adult / child > 12 y → 500 mcg (0.5 mL) · Adult / child > 12 y → 10 mg · Adult / child > 12 y → 200 mg · Child 6–12 y → 300 mcg (0.3 mL) · Child 6–12 y → 5 mg · Child 6–12 y → 100 mg · Child 6 m – 6 y → 2.5 mg · Child 6 m – 6 y → 50 mg · Child < 6 m → 250 mcg/kg · Child < 6 m → 25 mg
3. Fever and new confusion	1-A · 2-B · 3-D · 4-B · 5-B · 6-B
4. The sepsis bundle — in order	Top → bottom: 7 · 6 · 3 · 1 · 5 · 8 · 4 · 2
5. Chemo last week, now febrile	**C** — Treat as neutropenic sepsis now — e.g. piperacillin–tazobactam 4.5 g IV 8-hourly
6. BP 82/50 but HR 64	**D** — He takes a β-blocker
7. Clutching his neck while eating	**B** — 5 back blows, then 5 abdominal thrusts
8. Collapse on the ward — VF	1-B · 2-A · 3-C · 4-C · 5-C · 6-C
9. Pulseless — but the monitor shows a rhythm	1-D · 2-A · 3-A · 4-A · 5-B
10. Cardiac arrest drugs	Adrenaline → 1 mg IV every 3–5 min · Amiodarone → 300 mg IV → further 150 mg · Magnesium sulfate → 2 g (8 mmol) · Magnesium sulfate → Torsades de pointes · Calcium chloride 10% → 10 mL · Calcium chloride 10% → PEA with hyperK, hypoCa or CCB overdose · Sodium bicarbonate 8.4% → 50 mL (50 mmol) · Sodium bicarbonate 8.4% → Tricyclic overdose or hyperkalaemia
11. The relatives ask about the pupils	**C** — Prognosis is unreliable this early — rarely possible before 24–72 hr
12. Central line	**A** — Right internal jugular, high approach, USS-guided
13. Key numbers — Unit 1	qSOFA respiratory rate: **≥ 22 /min** · qSOFA systolic BP: **≤ 100 mmHg** · Lactate → repeat in 2 hr: **≥ 2 mmol/L** · Lactate → 30 mL/kg: **≥ 4 mmol/L** · Vasopressor MAP target: **≥ 65 mmHg** · Neutropenia threshold: **< 0.5 × 10^9/L** · Oliguria (adult): **< 50 mL/hr** · Shock: saline boluses: **20–30 mL/kg** · Adrenaline IM adult: **500 mcg (0.5 mL 1:1000)** · EpiPen dose: **300 mcg** · Glucagon on β-blocker: **1–2 mg every 5 min** · Anaphylaxis observation: **≥ 4–6 hr** · Compression depth / rate: **5–6 cm · 100–120/min** · Biphasic / monophasic shock: **150 J / 360 J** · Pads from a pacemaker: **> 15 cm** · Bicarbonate in arrest — consider if: **pH < 7.1, BE < −10** · Post-ROSC prognosis: **after 24–72 hr** · TTM temperature: **32–36 °C for 12–24 hr**
14. Viva — Unit 1	1. COPD chronic hypoxia gave high baseline scores → excessive O_2; also to identify sepsis early. 2. If hypotensive or lactate ≥ 4 mmol/L. 3. β-blockers, cardiac or neurological cause, or paradoxical bradycardia in haemorrhage. 4. Tamponade, massive PE, tension pneumothorax. 5. Usually no urticaria; treated with C1 esterase inhibitor. 6. Confirms tracheal tube placement and indirectly reflects cardiac output. 7. PHAT AI: Pneumothorax, Haemothorax, Arterial puncture, Thoracic duct damage, Air embolism, Infection. 8. Needle into vein → guidewire through needle → remove needle → dilator + cannula over wire → remove wire & dilator → aspirate, secure.

1. ECG basics	Paper speed: **25 mm/s** · 1 small / 1 large square: **0.04 s / 0.2 s** · Rate from the strip: **300 ÷ large squares in one R–R** · Normal axis: **–30° to +90°** · Normal PR: **0.12–0.2 s (3–5 small sq)** · Normal QRS width: **< 0.12 s (< 3 small sq)** · Normal QTc: **< 440 ms (QT/√RR)** · LVH voltage: **S V2 + R V5 > 35 mm** · Pathological Q: **> 0.04 s or > ½ the following R** · LBBB pattern: **W in V1, M in V6 — WiLLiaM** · RBBB pattern: **M (RsR′) in V1, W in V6 — MaRRoW**
2. Central crushing pain	1-B · 2-A · 3-D · 4-A · 5-C · 6-B
3. Name this ECG	**C** — Acute inferolateral STEMI
4. Name this ECG	**B** — Acute anteroseptal STEMI
5. Name this ECG	**D** — Pericarditis — widespread concave ST elevation
6. Pericarditis or STEMI?	1. STEMI · 2. Pericarditis · 3. STEMI · 4. Pericarditis · 5. Pericarditis · 6. STEMI · 7. Pericarditis · 8. STEMI
7. Normal ECG, normal troponin	**D** — Don't be reassured — serial troponins (and serial ECGs)
8. UA / NSTEMI — in order	Top → bottom: 6 · 5 · 1 · 4 · 2 · 3 · 7
9. Thrombolysis contraindications	1. Absolute · 2. Absolute · 3. Relative · 4. Relative · 5. Absolute · 6. Relative · 7. Relative · 8. Absolute · 9. Relative · 10. Absolute · 11. Absolute
10. Tenecteplase	**A** — 45 mg single IV bolus over 10 s
11. HR 34 and dizzy	1-D · 2-B · 3-C · 4-B
12. Which bradycardia risks asystole?	**B** — Sinus bradycardia in an athlete
13. Narrow regular tachycardia	1-A · 2-D · 3-C · 4-B
14. Broad complex, stable	**C** — VT — amiodarone 300 mg IV over 20–60 min, then 900 mg/24 hr
15. Name this rhythm	**A** — Torsades de pointes — $MgSO_4$ 2 g over 10 min
16. New AF	1-B · 2-D · 3-D · 4-B
17. Name this rhythm	**C** — Mobitz I (Wenckebach)
18. Name this rhythm	**A** — Mobitz II
19. Name this rhythm	**A** — Complete heart block
20. Name this rhythm	**B** — Paced rhythm
21. Tearing pain to the back	**A** — CT angiography; labetalol for BP; cardiothoracic early
22. Cocaine and BP 230/130	**D** — β-blockers
23. Key numbers — Unit 2	Aspirin in ACS: **300 mg** · Clopidogrel / prasugrel / ticagrelor: **300 / 60 / 180 mg** · Fondaparinux: **2.5 mg SC daily** · GTN infusion start: **0.6 mg/hr (if SBP > 90)** · PCI window / lysis deadline: **12 hr / 90 min** · Alteplase in STEMI: **15 mg → 0.75 mg/kg → 0.5 mg/kg** · Reteplase: **10 U + 10 U, 30 min apart** · Atropine: **500 mcg → max 3 mg** · Adrenaline infusion (brady): **2–10 mcg/min** · Pacing demand rate: **70/min** · Sync shock: broad / AF: **120–150 J** · Sync shock: flutter / SVT: **70–120 J** · Amiodarone infusion after load: **900 mg over 24 hr** · Hypertensive emergency BP: **> 230/130** · BP reduction first hour: **≤ 25%** · Keep K^+ post-MI: **≥ 4 mmol/L** · Dressler's: **2–14 weeks after 3% of MIs**

24. Viva — Unit 2	1. Pericarditis, ventricular aneurysm, Prinzmetal's, LVH, Brugada, HCM, benign early repolarisation. 2. Acute MI, hypothermia, hypocalcaemia, drugs (quinidine, TCAs), congenital (Romano–Ward). 3. Older patients, women, diabetics, heart failure — with LVF, syncope, confusion or stroke; higher mortality. 4. Pericarditis, large PE, sepsis; renal impairment reduces excretion. 5. V7–9, or mirror changes in V1–3: tall R, ST depression, upright T. 6. Concordant ST↑ > 1 mm · ST depression in V1–3 · discordant ST↑ > 5 mm. 7. It prolongs QT — give magnesium sulfate 2 g instead. 8. 2nd/3rd-degree heart block, severe hypotension, asthma. 9. Widened mediastinum, left effusion, trachea/NG deviated right, double knuckle, calcium sign > 5 mm. 10. Fever > 38 °C, ↑WCC, large effusion/tamponade, trauma, immunosuppression, oral anticoagulants, NSAID failure.

Respiratory & VTE · المناوبة ٣

1. Who gets which O_2 target?	1. High-flow (~100%) · 2. High-flow (~100%) · 3. 90–94% · 4. High-flow (~100%) · 5. High-flow (~100%) · 6. 88–92% · 7. 90–94% · 8. 88–92%
2. Will the cylinder last?	**C** — 45 minutes
3. House fire victim	**D** — COHb is read as oxyhaemoglobin — check COHb on a VBG
4. Type I or type II failure?	1. Type II · 2. Type I · 3. Type II · 4. Type II · 5. Type I
5. Acid–base at the bedside	1-D · 2-C · 3-A
6. Panic attack?	**C** — Exclude PE, DKA, poisoning — SpO_2, ECG, BMG, ABG if not settling
7. Can't finish her sentences	1-C · 2-D · 3-C · 4-B · 5-D · 6-C
8. Grade the asthma	1. Acute severe · 2. Life-threatening · 3. Acute severe · 4. Life-threatening · 5. Acute severe · 6. Acute severe · 7. Life-threatening · 8. Life-threatening · 9. Life-threatening
9. COPD, drowsy	1-D · 2-A · 3-A · 4-D · 5-C
10. Score the pneumonia	1-D · 2-B · 3-B
11. After the flu	**A** — Staphylococcus aureus
12. Sudden pleuritic pain	1-C · 2-B · 3-B · 4-C
13. Gasping, trachea deviated	**C** — Needle decompression 2nd ICS mid-clavicular line, then chest drain
14. Exudate or transudate?	1. Exudate · 2. Exudate · 3. Transudate · 4. Exudate · 5. Exudate · 6. Transudate · 7. Transudate · 8. Transudate · 9. Exudate
15. Pink frothy sputum	1-A · 2-D · 3-C
16. Post-op day 5, breathless	1-D · 2-A · 3-C · 4-D
17. Anticoagulants for PE	Rivaroxaban → 15 mg bd × 21 days, then 20 mg daily · Apixaban → 10 mg bd × 7 days, then 5 mg bd · Enoxaparin → 1 mg/kg SC bd (max 100 mg bd) · Dalteparin → 200 U/kg SC daily (max 18,000 U) · Unfractionated heparin → IV infusion — if eGFR < 20

18. Key numbers — Unit 3	Normal adult RR: **11–18 /min** · Hypoxia on air: **pO_2 < 10.6 kPa** · Hypercarbia: **pCO_2 > 6.0 kPa** · Anion gap normal: **12–16 mmol/L** · Osmolal gap normal: **< 10 mOsm/kg** · Lactate normal: **< 2.0 mmol/L** · Asthma SpO_2 target: **94–98%** · Prednisolone in asthma: **40–50 mg** · $MgSO_4$ in asthma: **1.2–2 g over 20 min** · Asthma ventilation: **Vt 6–8 mL/kg, RR 7–8, I:E 1:4** · COPD prednisolone: **30 mg × 7 days** · CPAP start: **5–8 cmH_2O** · Pneumothorax size cut-off: **2 cm at the hilum** · Aspiration needle / max volume: **16–18G / 2.5 L** · DVT likely (Wells): **≥ 2** · PE likely (Wells): **≥ 4.5** · Calf difference (Wells): **> 3 cm** · Effusion visible / clinical: **250 / > 500 mL**
19. Viva — Unit 3	1. CO poisoning (falsely high), poor perfusion/shock, hypothermia, methaemoglobinaemia, nail varnish, movement; it says nothing about pCO_2. 2. Lactic acidosis, ketoacidosis, renal failure, toxins (salicylates, methanol, ethylene glycol, metformin). 3. Needs ventilatory support or failing therapy: falling PEF, worsening hypoxia, hypercapnia, ↓pH, exhaustion, drowsiness, arrest. 4. Apnoea/arrest, coma, vomiting (unprotected airway), severe agitation, undrained pneumothorax. 5. Pulmonary oedema, infarction, vasculitis, aspergillosis, alveolitis, carcinoma, pancreatitis, subphrenic abscess. 6. Only if Wells ≤ 4 (PE unlikely) AND the D-dimer is normal. 7. ABCDE: Alveolar oedema (bat's wing), Kerley B lines, Cardiomegaly, Dilated upper lobe vessels, Effusions. 8. ARDS, SAH, fluid overload, hypoalbuminaemia, toxins/smoke, near drowning, high altitude, lymphangitis.

GI & Liver · ٤ المناوبة

1. Vomiting fresh blood	1-C · 2-D · 3-D · 4-A · 5-C
2. Variceal bleed — TAKES	Top → bottom: 5 · 1 · 4 · 3 · 2
3. Can he go home?	**B** — No — only a score of 0 can go home; > 0 needs urgent endoscopy
4. Glasgow–Blatchford points	Urea > 25 mmol/L → 6 · Hb < 100 g/L (man or woman) → 6 · SBP < 90 → 3 · Syncope → 2 · Pulse ≥ 100 → 1 · Melaena → 1
5. Fresh red blood PR, BP 90/60	**A** — An upper GI bleed
6. The PEG fell out 2 hr ago	**A** — Triage ahead, call gastroenterology; keep the track open (e.g. lubricated Foley ≤ original size, don't feed through it)
7. Hasn't eaten for a week	**D** — Phosphate
8. Where is the jaundice from?	1. Post-hepatic · 2. Intrahepatic · 3. Pre-hepatic · 4. Post-hepatic · 5. Post-hepatic · 6. Pre-hepatic · 7. Intrahepatic · 8. Intrahepatic · 9. Pre-hepatic
9. Read the LFTs	**A** — Obstructive (cholestatic)
10. Cirrhotic with fever and tender ascites	1-C · 2-C · 3-C
11. Key numbers — Unit 4	Blatchford to go home: **0** · Transfusion Hb target in UGIB: **> 70 g/L** · Terlipressin: **2 mg IV every 4–6 hr** · Omeprazole (known ulcer): **40 mg in 100 mL over 30 min** · Sengstaken oesophageal balloon: **30–40 mmHg** · Colorectal share of GI bleeds: **~20%** · LGIB cross-match: **4–6 units** · Refeeding risk: **not eaten ≥ 5 days** · Visible jaundice: **bilirubin > 51 µmol/L** · Refractory ascites 1-yr survival: **< 50%**

12. Viva — Unit 4	1. Only Glasgow–Blatchford 0 with no further bleeding; anyone > 0 needs urgent endoscopy. 2. Terlipressin 2 mg IV 4–6-hourly, prophylactic antibiotics, vitamin K if INR prolonged — then emergency endoscopy. 3. ↑ age, cancer, heart failure, liver disease, continued bleeding, ↑ urea, PR blood. 4. Obstructive: ALP rises more than ALT (with ↑GGT); hepatic: ALT predominant; AST:ALT > 1 suggests alcohol. 5. Not eaten ≥ 5 days, low BMI/anorexia, alcoholics.

Neurology · المناوبة ٥

1. Red flag or not?	1. Red flag · 2. Red flag · 3. Red flag · 4. Not a red flag · 5. Not a red flag · 6. Red flag · 7. Red flag
2. Thunderclap headache	1-C · 2-B · 3-A · 4-A
3. Hunt & Hess — 1 to 5	Top → bottom: 1 · 3 · 2 · 5 · 4
4. Triptan or not?	**A** — Hemiplegic migraine
5. Jaw pain when chewing	1-C · 2-A
6. Match the treatment	1. High-flow O_2 12 L/min · 2. Carbamazepine · 3. Heparin · 4. Blood patch
7. Right arm weak, can't speak	1-C · 2-C · 3-B · 4-A · 5-A
8. Exclusion or not?	1. Excludes · 2. Excludes · 3. Doesn't exclude · 4. Doesn't exclude · 5. Excludes · 6. Doesn't exclude · 7. Excludes · 8. Excludes · 9. Excludes
9. Ischaemic stroke, out of window	**A** — Swallow screen, then aspirin 300 mg PO/PR ASAP
10. Intracerebral haemorrhage	**B** — Andexanet · SBP < 140 within 1 hr
11. TIA — symptoms gone	**C** — Aspirin 300 mg now, TIA clinic < 24 hr, statin, no driving until seen
12. Status epilepticus — the ladder	Top → bottom: 1 · 3 · 5 · 2 · 4 · 6 · 7
13. No IV access	1-B · 2-C · 3-A
14. Can he go home?	**A** — Home: epilepsy clinic within a week, no driving/machinery, supervised bathing — documented
15. Unconscious, pinpoint pupils	1-C · 2-B · 3-A
16. Syncope — which need cardiology < 24 hr?	1. Cardiology < 24 hr · 2. Cardiology < 24 hr · 3. Classic vasovagal — can go · 4. Cardiology < 24 hr · 5. Cardiology < 24 hr · 6. Cardiology < 24 hr
17. UMN or LMN?	1. UMN · 2. UMN · 3. LMN · 4. LMN · 5. UMN · 6. LMN · 7. LMN
18. Ascending weakness	**D** — Serial vital capacity
19. Key numbers — Unit 5	Ottawa SAH age range: **15–39 y, peak within 1 hr** · CT sensitivity for SAH: **> 98% within 6 hr** · LP timing after onset: **> 12 hr** · Nimodipine: **60 mg PO 4-hourly** · Fever + headache antibiotic: **cefotaxime 2 g IV** · Sumatriptan: **50–100 mg PO or 6 mg SC** · Cluster O_2: **12 L/min, 15 min** · Stroke lysis BP limit: **< 180/105** · Alteplase stroke: **0.9 mg/kg (max 90), 10% bolus** · Platelets exclusion for lysis: **< 100** · ICH BP target: **SBP < 140 within 1 hr** · Status definition: **> 30 min** · Lorazepam in status: **4 mg IV, repeat once at 10 min** · Phenytoin: **20 mg/kg at 50 mg/min (max 2 g)** · Naloxone trial: **0.4–0.8 mg IV** · Stroke risk after TIA: **5% at 48 hr · up to 50% at 5 y**

20. Viva — Unit 5	1. After 6 hr or high suspicion — admit for LP > 12 hr after onset (xanthochromia). 2. To prevent/treat ischaemic deficits from vasospasm: 60 mg PO 4-hourly. 3. Unknown onset, seizure at onset, previous ICH, platelets < 100, stroke/head injury/surgery < 3 months, GI bleed < 3 weeks, DBP > 140. 4. Hypoglycaemia, Todd's paresis, hemiplegic migraine, meningitis/encephalitis, abscess, head injury, Bell's palsy, tumour. 5. Lorazepam 4 mg IV slowly, repeated once after 10 min (buccal midazolam 10 mg if no IV). 6. DWIM-CNOE: Drugs, Withdrawal, Infection, Metabolic, Cardiac, Neuro, Organ failure, Endocrine. 7. The ice pack test — cooling the eyelids improves ptosis. 8. Sudden memory loss, no other neuro signs, repetitive questioning, recovers within 24 hr, not a stroke predictor.

Endocrine & Electrolytes · المناوبة ٦

1. Found confused and sweaty	1-A · 2-D · 3-A · 4-B
2. Vomiting, deep sighing breaths	1-D · 2-C · 3-C · 4-D · 5-D · 6-B
3. DKA or HHS?	1. HHS · 2. DKA · 3. HHS · 4. HHS · 5. DKA · 6. HHS · 7. DKA · 8. DKA
4. Na 116, seizing	**B** — 2.7% saline up to 200 mL over 30 min, recheck
5. Speed limits	**D** — Chronic hyponatraemia ≤ 10 mmol/L per 24 hr · hypernatraemia ≤ 1 mmol/L per hr
6. Free water deficit	**C** — 4.2 L
7. K^+ is 7.4	**A** — Severe hyperkalaemia — peaked T, absent P, wide QRS
8. Hyperkalaemia — treatment order	Top → bottom: 4 · 3 · 5 · 2 · 1
9. Hyperkalaemia doses	Calcium chloride 10% → 10 mL IV over 5 min · Calcium chloride 10% → Stabilises membrane — doesn't lower K^+ · Actrapid + glucose → 10 U + 50 mL 50% glucose · Actrapid + glucose → ↓ K^+ up to 1 mmol/L in 1 hr, lasts 4 hr · Salbutamol → 5 mg neb · Furosemide (moderate) → 1 mg/kg IV
10. Calcium and digoxin	**D** — Over 30 min — hypercalcaemia potentiates digoxin toxicity
11. K^+ is 2.4	**A** — Prominent U waves — max 20 mmol/hr IV with cardiac monitoring
12. Shocked after stopping steroids	1-C · 2-D · 3-D
13. Thyroid storm	**D** — Aspirin — displaces thyroxine from TBG
14. Abdominal pain, dark urine on standing	**C** — IV diazepam
15. Key numbers — Unit 6	Normal glucose: **3.6–5.8 mmol/L** · Treat hypoglycaemia below: **3.0 mmol/L** · Glucagon: **1 mg SC/IM/IV** · DKA first saline: **1 L over 30–60 min, then 500 mL/hr × 2–3 hr** · DKA insulin: **0.1 U/kg/hr** · DKA: add 10% glucose when: **glucose < 14** · DKA KCl: **20 mmol/hr unless K^+ > 5.5** · HHS glucose: **> 30 mmol/L** · Hypertonic saline: **2.7% · 200 mL over 30 min** · Hyperkalaemia urgent: **> 6.5 mmol/L** · Hypokalaemia severe: **< 2.5 mmol/L** · Max K^+ infusion: **20 mmol/hr** · Addison / storm hydrocortisone: **100 mg IV** · Storm propranolol: **1 mg IV or 60 mg PO**

16. Viva — Unit 6	1. Total body K^+ is depleted; insulin shifts K^+ into cells → give 20 mmol/hr unless K^+ > 5.5. 2. When ketones < 0.3 mmol/L and pH > 7.3. 3. No — it stabilises the myocardium; insulin/glucose and salbutamol shift K^+ into cells. 4. Spurious (haemolysis), ↓ excretion (AKI, K-sparing), cell injury (rhabdo, burns, tumour lysis), shift (acidosis), ↓ aldosterone (Addison's, ACEi, NSAIDs). 5. Sudden withdrawal of long-term steroid therapy. 6. ↑K^+, ↓Na^+, ↓glucose, ↑urea, mild acidosis, ↑Ca^{2+}, eosinophilia.

Renal · V المناوبة

1. Does it meet AKI criteria?	1. Not (yet) AKI · 2. AKI · 3. Not (yet) AKI · 4. AKI · 5. AKI
2. Pre-renal, renal or post-renal?	1. Pre-renal · 2. Pre-renal · 3. Pre-renal · 4. Renal · 5. Post-renal · 6. Post-renal · 7. Post-renal · 8. Renal · 9. Renal
3. Not passing urine	1-C · 2-A · 3-A
4. AEIOU — indications for dialysis	Top → bottom: 5 · 2 · 1 · 3 · 4
5. Anuric, in pulmonary oedema	**A** — Urgent dialysis; meanwhile O_2 + nitrates
6. The fistula thrill is gone	**A** — Shunt thrombosis — vascular emergency
7. Cloudy PD bag	**C** — A surgical cause such as a diverticular abscess
8. Dysuria in pregnancy	1-A · 2-B
9. How long?	1. 7 days · 2. 3 days · 3. 7 days · 4. 5–10 days · 5. 5–10 days
10. Positive dipstick, no symptoms	**B** — No — asymptomatic bacteriuria in the elderly/catheterised isn't treated
11. Key numbers — Unit 7	AKI creatinine (48 hr): **≥ 26 µmol/L rise** · AKI creatinine (7 days): **≥ 50% rise** · AKI urine output: **< 0.5 mL/kg/hr for > 6 hr** · CKD problems start at GFR: **< 40 (esp. < 10)** · UTI threshold: **> 10^5 CFU/mL** · E. coli share of UTI: **90%** · Pyelonephritis temperature: **> 38 °C + loin pain** · Trimethoprim: **200 mg bd** · Nitrofurantoin: **50 mg qds**
12. Viva — Unit 7	1. Creatinine ↑ ≥ 26 µmol/L in 48 hr, or ≥ 50% in 7 days, or urine < 0.5 mL/kg/hr for > 6 hr. 2. AEIOU: Acidosis, Electrolytes (↑K^+), Intoxication, Overload, Uraemia. 3. Graft pain/swelling, ↓urine, fever, ↑creatinine — often infection; treat both and call the transplant team. 4. Trimethoprim in the 1st trimester; nitrofurantoin in the 3rd.

Haematology · Λ المناوبة

1. Platelets or clotting factors?	1. Platelet problem · 2. Coagulation factor problem · 3. Coagulation factor problem · 4. Platelet problem · 5. Platelet problem · 6. Coagulation factor problem
2. Reverse the anticoagulant	Warfarin → PCC (Beriplex) + vitamin K 10 mg IV · Dabigatran → Idarucizumab · Rivaroxaban / apixaban → Andexanet alfa · LMWH (within 24 hr) → Protamine sulfate
3. INR 8.2, no bleeding	**C** — Withhold warfarin; specialist advice before vitamin K_1 5 mg slow IV or PO
4. Starting warfarin	**B** — It is initially prothrombotic — cover with heparin/LMWH
5. Bleeding out	1-C · 2-D · 3-D · 4-B · 5-C · 6-D
6. What to request — and how long	Group & screen (electronic issue) → ~5 min · ABO/Rh-compatible → ~15 min · Full cross-match → Up to 1 hr · O Rh-negative → Immediately
7. Suspected transfusion reaction	Top → bottom: 1 · 4 · 5 · 3 · 6 · 2

8. Breathless after 2 units	**C** — TACO — transfusion-associated circulatory overload
9. Read the clotting	**D** — DIC — treat the cause, expert-guided products
10. Which product?	1. Cryoprecipitate · 2. PCC · 3. FFP
11. Sickle crisis	1-D · 2-D · 3-B
12. Key numbers — Unit 8	Thrombocytopenia / spontaneous bleeding: **< 100 / < 20 × 10^9/L** · Discuss with haem on anticoagulant if platelets: **< 50** · Tranexamic acid: **1 g IV within 3 hr** · FFP : red cells: **1 : 1** · Fibrinogen → cryo: **< 2.0 (keep > 1.5)** · INR/APTT target: **< 1.5× normal** · Red cells rise: **4 mL/kg → Hb ↑ ~10 g/L** · Red cells / TACO-risk rate: **90 min / 3.5 hr** · Platelets, FFP, cryo: **over 30 min** · Blood storage: **2–6 °C** · Sickle O_2 threshold: **SpO_2 < 94%** · HbS: trait / SS: **~40% / > 80%**
13. Viva — Unit 8	1. Platelet: immediate mucocutaneous bleeding, petechiae. Factor: delayed, deep bleeding into joints and muscles. 2. Low platelets and fibrinogen, raised INR, APTT and FDPs. 3. Prothrombin complex concentrate plus vitamin K 10 mg IV. 4. 1:1 FFP:red cells; platelets > 75; fibrinogen > 1.5 (cryo below 2.0); INR/APTT < 1.5×; tranexamic acid 1 g within 3 hr. 5. Hypothermia, citrate toxicity (↓Ca^{2+}, ↓Mg), ↑K^+ from stored blood (then transient ↓K^+ at 24 hr). 6. Acute chest syndrome — chest pain, hypoxia, new pulmonary infiltrates.